

Eliminating the Cruise-Efficiency Penalty of Hybrid VTOL: The meryemAircraft Tail-Sitting Blended-Wing-Body Configuration with Propeller-Only Control

Meryem Gülmen, Berke Gülmen, Ömer Gülmen

Abstract. Hybrid vertical take-off and landing (VTOL) aircraft combine runway independence with wing-borne cruise, but purchase that combination at a cost to cruise efficiency. This paper argues the cost is architectural rather than a defect of implementation. It is charged in three currencies — the mass of hover hardware carried through cruise, its drag when exposed in cruise, and a power system sized by a condition holding for roughly two percent of the flight — and every known remedy reduces one currency by increasing another. Stating the cost this way makes its escape condition explicit: it is charged whenever hover and cruise are served by hardware that is not the same hardware, in the same orientation, doing the same job. A configuration satisfying that condition is proposed — an uncrewed tail-sitting blended-wing body in which one coaxial counter-rotating pair at the nose produces all thrust in both regimes, four small coaxial pairs at the wing tips produce attitude moments only, and a deployable strip in the nose-propeller slipstream supplies the rolling moment that body-axis-parallel thrust vectors cannot generate. The aircraft has no control surfaces, no tilting or retraction mechanism and no dedicated lift system. Two reference designs are sized twenty times apart in mass, at 50 kg and 1000 kg, from identical equations, with the governing fractions preserved across that range. Two findings changed the study: the tip frames must be faired, and transition altitude loss falls with rotation time rather than rising with it. The study is largely analytical, with no wind-tunnel or flight validation, and the mass budget is a target. The two aerodynamic coefficients that carry the most weight are not replaced by computation but bounded by it — the zero-lift drag by a three-dimensional solution reported with a measured uncertainty budget.

Keywords: vertical take-off and landing; tail-sitter; blended wing body; uncrewed aerial vehicle; series hybrid propulsion; cruise efficiency; aircraft configuration design

1. Introduction

Powered flight for uncrewed aircraft is dominated by two configuration families, and each is bounded by a different limit.

Fixed-wing aircraft carry payload efficiently over long distances because the wing sustains the vehicle without continuously expending power on lift. Their limit is not aerodynamic but infrastructural: they require a runway, a catapult, or an equivalent launch and recovery installation. That requirement is expensive, fixed in place, and scales poorly — a larger fixed-wing aircraft demands a longer runway, stronger pavement, and wider taxiways, so its growth is gated by the ground rather than by the air.

Rotary-wing and multirotor aircraft remove that requirement entirely. They take off and land vertically, hover, and operate from confined sites. Their limit is the converse: without a wing, every second of flight is paid for with installed power, so range and endurance remain modest and degrade further as the vehicle grows. Figure 1 places the two families against the two capabilities and marks the corner that neither occupies.

Two configuration families and the limit of each

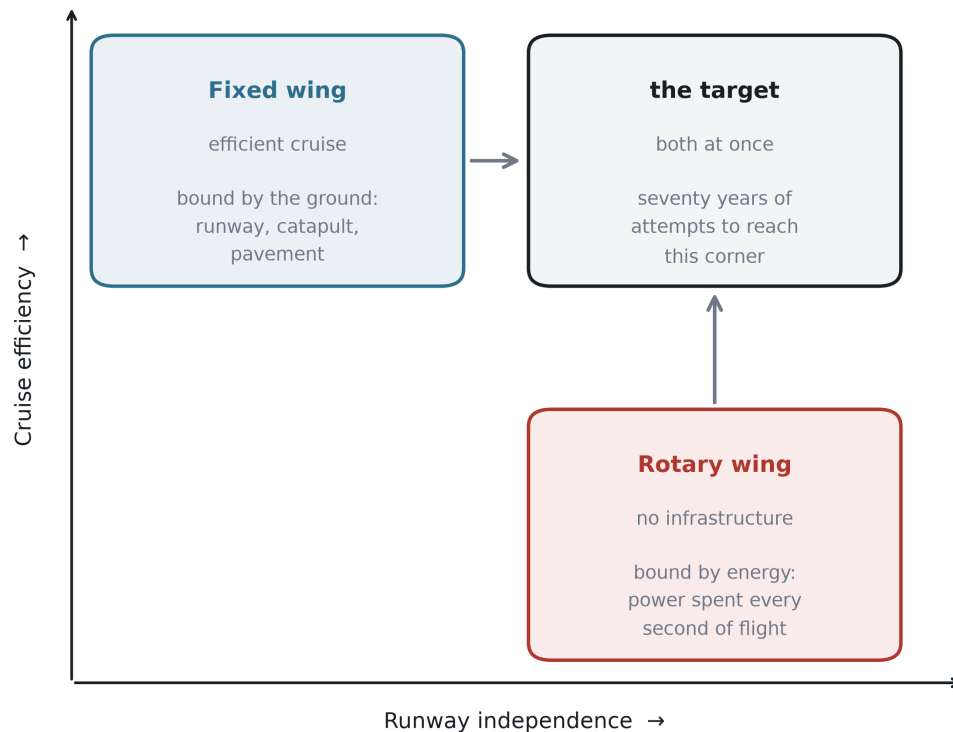


Figure 1. Two configuration families and the limit of each. Fixed-wing aircraft occupy the efficient-cruise corner and are bounded by ground infrastructure; rotary-wing aircraft occupy the runway-independent corner and are bounded by installed power. Seventy years of hybrid development has been an attempt to reach the corner that neither family occupies.

The demand to combine the two has been continuous and expensive. Tail-sitting prototypes were flown in the 1950s, vectored-thrust and tilt-wing aircraft in the 1960s, tilt-rotors from the 1980s, and a broad family of hybrid vertical take-off and landing (VTOL) uncrewed aircraft since the 2010s. Different nations, services and propulsion philosophies have attacked the same problem for seventy years. No field sustains that level of effort against a need that is not real.

Contemporary hybrid VTOL aircraft do combine the two capabilities, and several are in service. This paper does not dispute that. It argues, instead, that they purchase the capability at a specific and quantifiable price, and that the price is charged to cruise efficiency.

The most common hybrid architecture — separate lifting rotors for the vertical phase and a separate propulsor for cruise — carries its hover hardware for the whole flight while using it for a few percent of it. The consequences are measurable. Lift propellers left exposed in forward flight impose a parasitic drag penalty large enough that retracting

them recovers a substantial fraction of it; rotor-wake interference raises drag further in the hybrid regime; and the second propulsion system remains as dead mass for the entire cruise. Tilting architectures avoid the dead mass but substitute mechanical complexity, gyroscopic coupling during transition, and a non-trivial transition control problem.

The central observation of this paper is that these penalties are not defects of implementation. They arise from the architecture itself: from the decision to provide hover and cruise with different hardware. Better engineering can move the penalty between mass, drag and powertrain sizing, and can reduce any one of them, but it cannot remove the trade, because the trade is structural.

This paper presents a configuration that does not make that trade. In the proposed arrangement the aircraft sits on its tail, its entire body is a lifting blended-wing body, thrust for every flight phase is produced by a single coaxial counter-rotating propeller pair at the nose, and attitude control is produced by four small coaxial pairs mounted on rigid frames at the wing tips. The aircraft has no elevons, no rudder, no tilting mechanism and no dedicated lift system. Roll authority, which cannot be generated by thrust vectors parallel to the body axis, is provided by a level-controlled deployable strip on the lower surface positioned within the main propeller slipstream, so that it remains effective at zero airspeed.

Because hover and cruise are served by the same hardware, none of the three penalties is incurred. The cost that is incurred — the mass and drag of the control propellers and their supporting frames — is reported and quantified rather than omitted.

Contributions. This paper (i) states the cruise-efficiency penalty of hybrid VTOL as an architectural rather than an implementation property, and supports that statement with published figures; (ii) describes a configuration in which the penalty does not arise; (iii) sizes two reference designs, at 50 kg and 1000 kg maximum take-off weight, from first principles; and (iv) shows which properties of the configuration are preserved across that twenty-fold mass range — disc loading, energy-buffer mass fraction and tip-frame drag fraction — and identifies the two that are not: transition duration, and the ratio of propeller diameter to span.

Scope. This is a configuration study. It contains no wind-tunnel measurement and no flight test. Its numerical results are analytical estimates from stated assumptions, with one exception: the zero-lift drag of the wing and centre body has been computed three-dimensionally, and Section 6.6 reports it with a measured uncertainty budget. The mass budget is a target rather than a finding. Section 8 states these limitations explicitly.

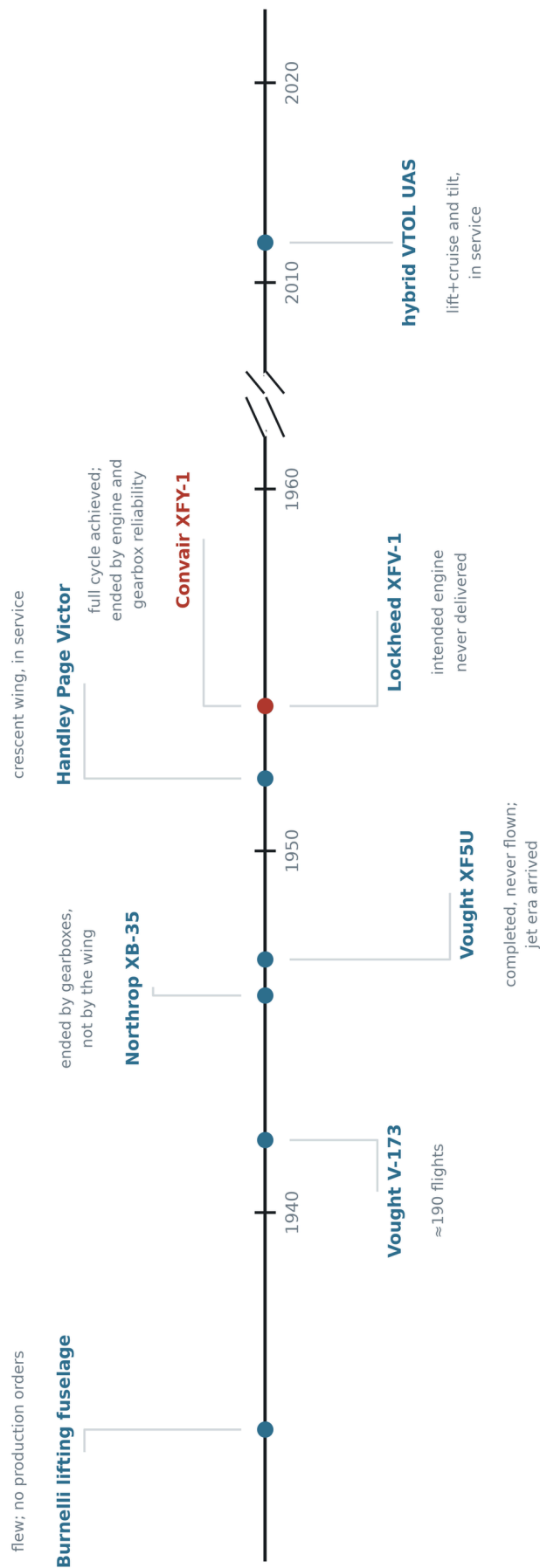
The remainder of the paper is organised as follows. Section 2 reviews seventy years of attempts to merge the two configuration families and argues, on the evidence of two contemporary NASA reviews, that they ended for reasons external to the configuration — principally engine and transmission reliability — while the one difficulty those reviews document most fully, the workload of a pilot flying a vertical descent, is also the one that an uncrewed aircraft removes outright. Section 3 sets out the architectural tax in its three currencies, shows that architectural remedies transfer it rather than remove it, and derives the condition under which it would not be charged. Section 4 describes the proposed configuration, which is built to satisfy that condition. Section 5 audits the claim

bill by bill and states what the configuration does pay. Section 6 sizes two reference designs twenty times apart in mass and examines how the proportions scale. Section 7 treats the flight profile and the transition manoeuvre, including a result that contradicts a common assumption about how quickly a tail-sitter should rotate. Section 8 states the limitations, and Section 9 concludes.

2. Background: seventy years of attempts

The configuration proposed in this paper is new, but the problem it addresses is not, and neither are several of its ingredients. This section reviews the attempts that preceded it. The purpose is not to establish priority but to establish two things: that the need has been pursued continuously for seventy years, and that the pursuit was rarely abandoned because the aerodynamics failed. Figure 2 places the programmes discussed below on a single timeline, with the recorded reason each one stopped.

Seventy years of attempts to merge the two families



Marked in red: the one programme that completed the full cycle — vertical take-off, transition, cruise, transition, vertical landing.
Two contemporary NASA reviews record that its testing was curtailed by engine and gear-box reliability, not by the configuration.

Figure 2. Seventy years of attempts to merge the two families, with the recorded reason each programme ended. The two entries of 1954 and the reason the XFY-1 programme stopped are taken from [1] and [2]; the remaining entries carry no numerical claim.

2.1 Removing the fuselage

The idea that a transport aircraft should carry its payload inside a lifting surface rather than inside a cylinder is as old as the transport aircraft itself. Burnelli's lifting-fuselage designs, flown in a succession of prototypes from the 1920s to the 1940s, placed cabin and cargo inside a thick aerofoil-shaped centre body that contributed lift instead of only drag. The aircraft flew, repeatedly and over two decades. They never entered series production. The reasons for that are outside the scope of this paper and are disputed; what matters here is only that the layout was flown rather than merely proposed, and was not abandoned at the drawing board.

The Northrop XB-35 carried the same idea to its limit: a bomber with no fuselage and no tail at all. Its programme is often cited as evidence that the flying wing was premature. Whatever weight that reading deserves, the aircraft's best-documented difficulties lay in its power transmission rather than in its aerodynamics: the contra-rotating propellers were driven through remote gearboxes and long extension shafts, and were eventually changed to single-rotation units. The distinction between an airframe and the machinery installed in it matters for the present work, and Section 4.3 returns to it — the configuration proposed here uses counter-rotating propellers but never builds a contra-rotating gearbox, because each rotor of a pair is driven by its own electric machine on a common axis.

Vought's V-173 and XF5U pursued the opposite extreme of the same intuition — a wing of very low aspect ratio with large propellers at the tips, intended to work against the tip vortices. The V-173 flew roughly two hundred times and supported the low-speed claims made for it. The XF5U was completed and never flown; the programme ended as the services moved to jet propulsion.

The recurring pattern is worth stating plainly: in each case the aircraft flew, the configuration was not disqualified in flight, and the programme ended for a reason that came from outside the configuration.

2.2 Standing the aircraft on its tail

The tail-sitter is the most direct answer to the runway. If the aircraft can point its thrust line at the ground, it needs no separate lift system, no tilting mechanism and no second propulsion group — the same propeller that cruises also hovers. Two American prototypes flew this idea in 1954.

The Lockheed XFV-1 never completed the cycle it was built for. Two contemporary NASA reviews of United States V/STOL development record what it did instead: its "highly tapered, straight-wing design made the transition to vertical flight only at altitude, using a jury-rigged, landing-gear cradle for conventional takeoff and landings" [1,2]. It was cancelled with the question it was built to answer still unanswered.

The Convair XFY-1 did answer it. It flew vertically in August 1954, and "six transitions to conventional flight were successfully completed" [2]. The concept was demonstrated.

What the aircraft was like to fly, and why the programme stopped, are recorded in the same two reviews — one of which states that most of its content came from the author's first-hand flight-test experience [1]. Their assessment is more interesting than the account usually given.

The configuration itself is judged favourably: "good configuration arrangement for low- and high-speed compatibility", with a high-speed potential of about 500 mph [1]. What is judged poorly is the machinery and the cockpit around it — "poor mechanical control system features including low actuator response rate"; "difficult to hover precisely over a spot"; "tip-over tendencies noted when on ground in gusty air"; "gust sensitivity bothersome to pilot during takeoff and landing phases" [1]. Of the landing itself, the second review is explicit about the cause: the pilot skill required was driven by "the unusual spatial orientation where the pilot looked over his shoulder and down", by "the sensitivity to atmospheric turbulence", and by "reduced control power near touchdown" [2]. The precision of flightpath control these concepts offered "was, needless to say, less than desired" [2].

And then the reason the programme ended, stated the same way in both documents:

*"Six transitions to conventional flight were successfully completed **before testing was curtailed because of engine and gear-box reliability problems.**" [2]*

That sentence corrects the account usually given — including an earlier draft of this paper. The XFY-1 was not stopped by the pilot workload. The workload was real, separately documented and severe, but what curtailed the testing was mechanical reliability in the engine and the gearbox. The tail-sitter of 1954 therefore shares its cause of death with the XB-35 and with the XFY-1's undelivered engine: **the powerplant and its transmission, not the configuration.**

Two further findings from these reviews bear directly on the present work, and both cut against the usual summary.

The first is what NASA identifies as the *foremost* deficiency of the tail-sitter concept — not the landing, but the absence of a short take-off option: "foremost among the deficiencies was the lack of STOL operational capability which could improve the poor payload and range capabilities of these aircraft" [2]. That criticism is aimed at an aircraft whose payload and range were poor for other reasons, and it does not transfer automatically to a configuration designed for cruise efficiency; but it is the judgement of the reviewers and it is recorded here rather than omitted.

The second is a positive finding that is rarely quoted: "dispensing with a conventional landing gear improved the empty weight fraction for these VATOL aircraft" [2]. Standing on the tail removed a mass item rather than adding one. Section 4.5 returns to this, because the present configuration takes the same benefit further — its landing structure is also its control moment arm.

Finally, the pilot workload matters for a reason that has nothing to do with blame. It is the one item on the list that an uncrewed aircraft removes outright. "Unusual spatial orientation where the pilot looked over his shoulder and down" is a statement about a human being in a cockpit. A vehicle whose attitude comes from an inertial measurement

unit and whose height above ground is a sensor reading has no unusual spatial orientation, because it has no orientation to be disoriented in.

The rest of the list does transfer, and this paper does not pretend otherwise. Tip-over tendencies in gusty ground wind, difficulty holding a precise hover, sensitivity to turbulence and reduced control power near touchdown are properties of standing an aircraft on its tail, not of having a pilot. Section 8 treats them as inherited.

One further detail from the record is worth noting because Section 7 arrives at the same place from theory. The XFY-1's return transition was flown as "a zoom climb... to achieve a vertical attitude for the descent and to reduce airspeed (altitude gain of about 3,000 ft)" [2]. The 1954 aircraft climbed into its transition. So, for different reasons and by a different argument, does the aircraft proposed here.

2.3 Distributing sweep along the span

The proposed planform varies its leading-edge sweep continuously from root to tip while holding the trailing edge at a constant angle. The principle of treating sweep, chord and thickness as one coupled distribution rather than three independent choices is not new; the crescent wing of the Handley Page Victor is its best-known expression, its sweep decreasing outboard so that the wing was not governed by its most vulnerable station.

The present aircraft is subsonic and does not inherit the transonic motivation that produced that planform. What it takes is the structural idea alone, and it takes it in a much reduced form: as Section 4.2 records, the sweep variation actually realised here is under seven degrees. No claim of descent from the crescent wing is made, and none is needed. Section 4.2 states the values used, and Section 8 states plainly that they were chosen rather than derived.

2.4 The contemporary hybrids

The problem did not go away when the prototypes did. Since roughly 2010 a large family of hybrid VTOL uncrewed aircraft has reached service, in two dominant architectures. Lift-plus-cruise aircraft carry a set of rotors for the vertical phase and a separate propulsor for cruise, and fly as a fixed-wing aircraft in between. Tilting architectures — tilt-rotor, tilt-wing and tilt-nacelle — reuse the same propulsors in both regimes by rotating them.

Both work. Both are in operational use. This paper does not claim otherwise, and Section 6.5 places the proposed reference designs alongside them without ranking them. What Section 3 argues is that each of these architectures pays for its capability in a way the other does not, that the payment can be moved between mass, drag and power system sizing, and that it cannot be brought to zero as long as hover and cruise are served by different hardware or by hardware that must move between two roles.

2.5 What this history does and does not show

It would be too convenient to declare that every one of these programmes ended for reasons external to its configuration. Some of the difficulties were real and internal, and

this paper inherits them. The tail-sitter's vertical descent is genuinely harder than a runway landing. A tail-sitting aircraft is more exposed to crosswind on the ground than a conventional one. And a set of propellers whose thrust vectors are all parallel to the body axis cannot, by construction, produce a rolling moment — a limitation that applies to this configuration exactly as it applied to its predecessors, and which Section 4.4 addresses rather than avoids.

What the history does show is that the concept was never given a fair verdict under present-day conditions. The programmes of the 1950s were closed by engine deliveries, by jet-era procurement priorities, and — in the case where two contemporary NASA reviews record the reason directly and identically — by engine and gearbox reliability [1,2]. Not one of them was closed because the configuration failed to fly; one of the reviews rates the configuration itself favourably while condemning the machinery around it [1]. The human pilot, whose difficulties are the best-documented part of the record, is the single constraint that an uncrewed aircraft removes entirely. What is available now that was not available then — electric drive on each rotor, sensor-based attitude reference, and enough onboard computation that stability need not come from the airframe alone — removes exactly the obstacles that stopped it.

What the history does not settle is why the contemporary aircraft that did succeed still pay for their vertical capability, and what exactly they pay. That is the subject of the next section.

3. The architectural tax of hybrid VTOL

Hybrid VTOL aircraft work. The argument of this section is not that they do not, but that they pay for the capability in a way that can be located precisely, that the payment appears in three different currencies, and that an improvement in one currency is usually a transfer into another rather than a reduction. The section closes by stating the condition under which the payment would be zero — a condition none of the current architectures satisfies, and which Section 4 is built to satisfy.

3.1 The root: a duty cycle that does not match the hardware

Every hybrid VTOL aircraft contains hardware whose only purpose is the vertical phase. That phase is short. For a mission of one hour, a take-off, a transition, a return transition and a landing occupy on the order of a minute — roughly two percent of the flight. The remaining ninety-eight percent is spent carrying that hardware through the air.

This is not an implementation defect and it cannot be engineered away by making the hardware better, because it is a statement about duty cycle rather than about quality. A lighter lift rotor is still carried for the whole flight. A cleaner lift rotor is still carried for the whole flight. The mismatch between how long a component is needed and how long it is present is the origin of all three bills below.

3.2 Bill 1 — mass

The most direct payment is dead mass. A lift-plus-cruise aircraft carries two propulsion groups: rotors, motors, mounts, wiring and structural reinforcement for the vertical phase, and a separate propulsor with its own installation for cruise. The vertical group is inert for the whole cruise, but it is still lifted, and lifting it consumes energy in proportion to its weight and inversely to the lift-to-drag ratio.

Its cost is also not linear, because mass growth feeds itself. Writing the maximum take-off mass in terms of the payload and the empty and energy fractions,

$$MTOW = m_{\text{payload}} / (1 - f_{\text{empty}} - f_{\text{energy}})$$

shows that additional empty mass does not add to MTOW once, but through a multiplier that grows as the denominator shrinks. In the vertical phase the same increment is charged again, because hover power scales with weight to the three-halves power:

$$P_{\text{hover}} = W^{1.5} / (\eta \sqrt{2\rho A})$$

so a mass increment raises the hover power requirement faster than proportionally, which raises installed power, which raises mass. This is the mechanism by which a modest dead-mass fraction becomes a large payload penalty.

This bill is not hypothetical, and it has been identified independently. In a NASA study that sized four VTOL architectures against a common mission with a common set of tools, the lift-plus-cruise concepts came out as the heaviest of the vehicles examined, and the authors are explicit about the cause:

"The weight of the Lift+Cruise concepts is heavier in general than for the other vehicles. This is not driven by the cruise power draw, as the L/D_e of the Lift+Cruise is indeed higher than the other vehicles. Hover power is higher, but the most likely targets for reducing vehicle weight are the extra empty weight items on board in hover (wing and propeller)."

A second NASA review of United States V/STOL development states the structural half of the same bill as a general principle, drawn from the failure of a tilt-prop aircraft whose propeller separated in flight after a gearbox mounting fatigued: "this exemplified an inherent deficiency of this VTOL (lift) arrangement: **to safely transmit power to the extremities of the planform, very strong (and fatigue-resistant) structures must be incorporated with an obvious weight penalty**" [2].

Distributing lift or thrust across the span is therefore not only a matter of carrying rotors and mounts. It obliges the structure that reaches them to be strong enough to transmit power to the planform extremities and fatigue-resistant enough to keep doing so. That obligation is charged to mass, and it is charged whether or not the distributed propulsors are running.

The finding is worth reading carefully, because it separates the two things this paper is at pains to separate. The lift-plus-cruise vehicle is *aerodynamically better* than the

alternatives it was compared against — its cruise efficiency is higher, and the study says so. It is nevertheless the heaviest, and the reason given is the hardware it carries in order to hover. That is Bill 1, stated by an independent source in its own terms: not a failure of engineering, but the cost of an architecture.

3.3 Bill 2 — drag

The second payment is aerodynamic and is charged only to those architectures that leave hover hardware exposed in forward flight. Rotors stopped in the airstream, the booms that carry them, and the interference between their wakes and the wing all add drag in the regime where the aircraft spends nearly all of its time.

The cleanest available measurement of this bill is a controlled comparison within a single aircraft. In a doctoral study, one uncrewed airframe was tested in a wind tunnel in four configurations: clean, with the vertical-lift motors and their supporting beams installed and the propellers left free to align with the flow, with the same hardware but the propellers held perpendicular to the flow, and with the propellers retracted into the fuselage. The maximum lift-to-drag ratios measured were:

Configuration	Maximum L/D
Clean airframe, no hover hardware	≈ 17
Hover hardware installed, propellers aligned with the flow	≈ 13
Hover hardware installed, propellers perpendicular to the flow	≈ 9

Two numbers follow, and both are measured rather than estimated. Installing the hover hardware costs about a quarter of the aircraft's lift-to-drag ratio. Failing to let the propellers align with the flow costs about a third of what remains. A second model built on a different airframe reproduced the pattern, at $L/D \approx 11$ with the propellers retracted against ≈ 8 with them exposed.

Because range is linear in lift-to-drag ratio for a fixed energy system, this ladder translates directly into range; Figure 12 carries it onto the range axis for both a battery-electric and a fuel-burning energy system.

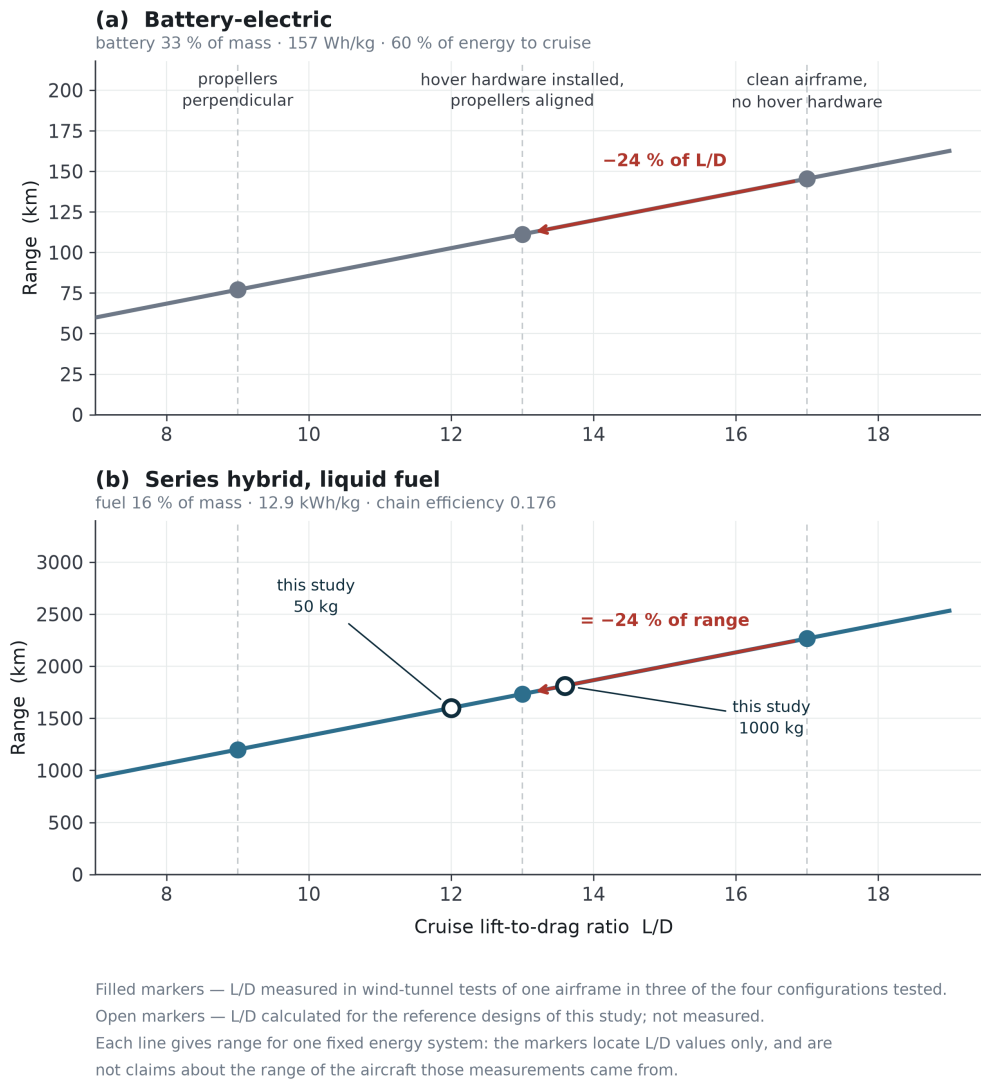


Figure 12. Range against cruise lift-to-drag ratio for (a) a battery-electric and (b) a series-hybrid energy system. Filled markers are L/D values measured in wind-tunnel tests of one airframe [3]; open markers are the calculated reference designs of this study.

Expressed as drag rather than efficiency, retracting the propellers reduced drag by 34 % relative to the standard quadplane configuration on one model and by 30 % on the other. The author of that study is careful about which comparison is legitimate: measuring the retracted aircraft against *itself* with the propellers deployed gives 63 %, and he explicitly rejects that figure in favour of the comparison against a conventional quadplane. The caution is worth adopting.

A third finding from the same tests matters more than either number, because it constrains what can be done about the penalty. The drag is not dominated by the propeller blades:

"The difference between propellers parallel to the airflow and without propellers is modest. The drag produced by the motors is significant."

The bill is charged mainly by the motors and the beams that carry them — hardware that cannot be feathered, folded or aligned away, because its cost is its presence.

Two further measurements support the direction of this result without being combined with it. Wind-tunnel characterisation of a QuadPlane uncrewed aircraft found that the highest lift and the least drag occurred in fixed-wing mode at both cruise airspeeds, that drag in the hybrid regime exceeded drag in either pure mode because of adverse flow interactions, and — a point that matters for how such aircraft are designed — that a simulation model assuming negligible interaction between the rotors and the structure "always predicts higher lift and lower drag than were experimentally observed."

Separately, a wind-tunnel study of twenty-six stationary lift propellers held edge-on to the flow found that their drag scales with frontal area and with the square of airspeed, that blade pitch adds to it, and that the hover powertrain components "added a significant amount of aerodynamic drag during forward flight" in the absence of a mechanism to stow them.

The important property of this bill is not its size but where it is charged. It is charged per unit time in cruise, so it grows with exactly the quantity the aircraft exists to maximise.

3.4 Bill 3 — power system sizing

The third payment is the least visible and often the largest. A VTOL aircraft must install enough power to hover, but it only uses that much power for the two percent of the flight in which it hovers. The ratio between the two demands follows from the governing equations rather than from any design choice. Taking hover power from momentum theory and cruise power from the drag polar,

$$\begin{aligned} P_{\text{hover}} / W &= \sqrt{(DL / 2\rho)} / \eta_h & (DL = W/A, \text{ disc loading}) \\ P_{\text{cruise}} / W &= V / ((L/D) \eta_p) \end{aligned}$$

so that

$$P_{\text{hover}} / P_{\text{cruise}} = \sqrt{(DL / 2\rho)} \cdot (L/D) / V \cdot (\eta_p / \eta_h)$$

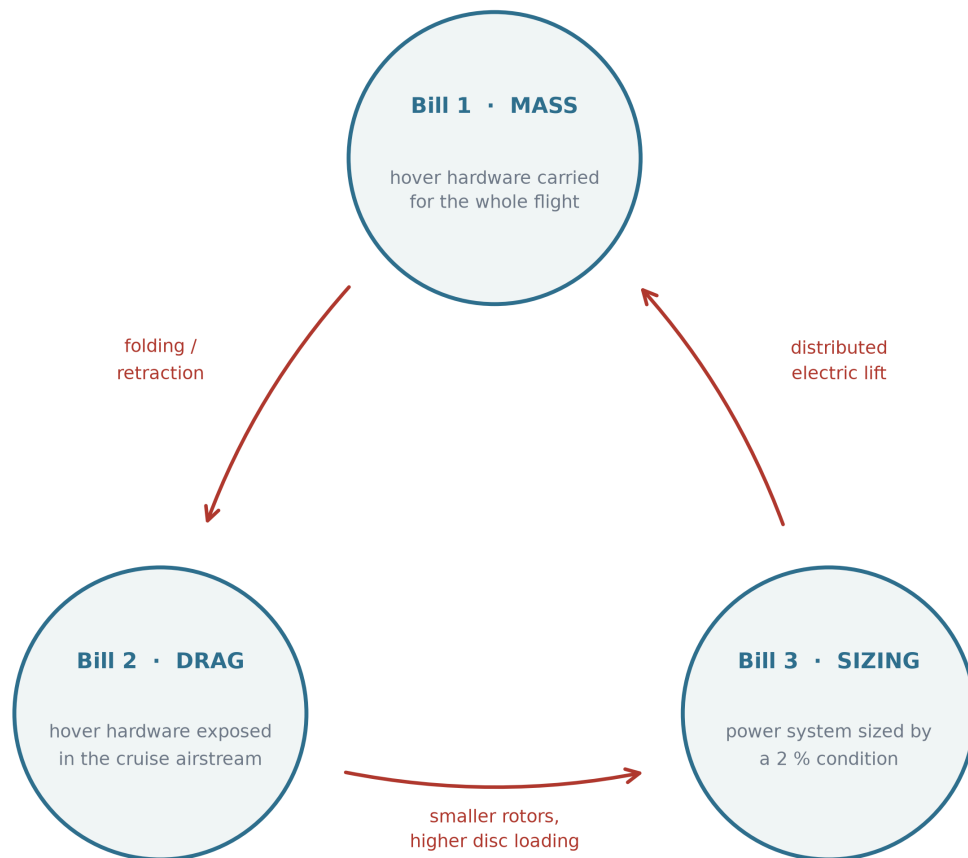
Every term on the right is a property of the configuration, not of the workmanship. A vehicle with a disc loading of 100 N m^{-2} , a cruise lift-to-drag ratio of 15 and a cruise speed of 30 m s^{-1} needs roughly four times as much power to hover as to cruise; raising the disc loading raises the ratio as its square root. The prediction is borne out in flight: a carbon-fibre tail-sitter reported in the literature measures its level-flight power consumption at one fifth of its hover power, which is the same ratio this expression gives for an aircraft of that class. The power system is therefore sized by a condition that holds for a minute and is then carried, unused, for an hour.

The consequence propagates. Sizing by hover means an oversized engine, or a battery that must deliver a peak it will rarely be asked for, or both — and whichever of the two is chosen, the extra installed capacity is mass, which returns to Bill 1.

3.5 The bills are one quantity in three currencies

The three bills are not independent problems with three independent fixes. Each known architectural move reduces one and increases another. Figure 3 shows the three bills and the moves that convert one into another; Table 1 lists the same moves in full.

Three bills — and the moves that convert one into another



Every known architectural remedy reduces one bill by increasing another.

The measured instance: removing 30 % of drag at a cost of 5 % of mass moved the range by under 2 %.

Figure 3. The three bills of the architectural tax and the moves that convert one into another. Every architectural remedy surveyed in Table 1 reduces one bill by increasing another. The measured instance quoted at the foot of the figure is derived in Section 3.5 from [3].

Table 1. Architectural moves and the bills they transfer.

Move	Bill it attacks	Bill it creates
Distributed electric lift rotors	3 — the cruise engine no longer sizes to hover	1 and 2 — many rotors and mounts, permanently carried and exposed
Folding or retracting lift rotors	2 — the exposed rotor is removed from cruise	1 — mechanism, actuation, locking, and a new failure mode
Tilt-rotor, tilt-wing, tilt-nacelle	1 — one propulsion group serves both regimes	mechanical complexity, gyroscopic coupling during rotation, and a transition control problem
Higher disc loading, smaller rotors	1 and 2 — smaller, lighter, cleaner rotors	3 — hover power rises with $\sqrt{DL}$
Lower disc loading, larger rotors	3 — hover power falls	1 and 2 — larger structure, larger exposed area

The table is not only an argument. One of its rows has been measured. In the study cited in Section 3.3, the retraction system that removed thirty percent of the drag was then costed: applied to a passenger eVTOL of known characteristics, with the retraction mechanism assessed at five percent of vehicle mass, the maximum range rose from 119 km to 121 km — an improvement of under two percent. The drag was genuinely removed and the range barely moved, because the mechanism that removed it was itself carried.

That is the transfer in Table 1, observed rather than asserted: Bill 2 was paid off by borrowing from Bill 1, and the balance was very nearly unchanged. What did improve was speed — the airspeed for maximum range rose by 5 m s^{-1} , and an 80 km mission could be flown 10 m s^{-1} faster — which is a real operational gain, and one worth having, but it is not a reduction of the tax. It is a change in the currency in which the tax is returned.

The same study notes that for a surveillance aircraft, whose endurance is maximised at low airspeed where the drag reduction is least effective, even that gain largely disappears.

Read as a whole, the table describes a pattern rather than a law. Across every move listed, the cost of giving a wing-borne aircraft a vertical capability behaves as though it were conserved: architectures do not remove that cost, they choose the currency in which to pay it.

It should be said plainly that nothing in physics requires this. No conservation principle is being invoked, and an architecture that reduced all three bills at once would be a genuine contribution rather than a contradiction. The claim here is empirical and bounded: among the architectures surveyed, none does, and Section 3.3 supplies a measured instance of the transfer rather than an assumed one. This is why seventy years of engineering effort has improved hybrid VTOL aircraft considerably without producing one whose cruise efficiency matches a comparable fixed-wing aircraft.

3.6 The condition for a zero bill

Stating the tax this way makes its escape condition explicit. The bill exists because hover and cruise are served by hardware that is not the same hardware, doing the same job, in

the same orientation. Relax any part of that and a bill appears:

- **Different hardware** → Bills 1 and 2. The unused set is carried and, if exposed, drags.
- **Same hardware, different orientation** → the tilting family. The mechanism that changes the orientation is itself mass, complexity and a control problem.
- **Same hardware, same orientation, different sizing point** → Bill 3, unless the hover peak is supplied from somewhere other than the continuous power source.

The condition for paying nothing is therefore that the same propulsors, fixed in the same orientation relative to the airframe, produce both the hover thrust and the cruise thrust — with the aircraft itself changing orientation rather than any part of it — and that the difference between the hover peak and the cruise demand is supplied by a buffer rather than by permanently installed continuous power.

That is a description of a tail-sitter with a buffered series-hybrid powertrain. It is also, precisely, the configuration described in Section 4.

3.7 Why the market looks the way it does

One observable consequence supports the argument, and it has been stated independently. Surveying the field, the study cited above concludes that multirotors are efficient in hover and suited to short-range missions, that vectored-thrust aircraft are efficient in cruise and suited to long-range missions, and that "lift plus cruise eVTOLs are a compromise, but they are slowed down by the drag of the lift propellers."

Hybrid VTOL aircraft occupy a narrow band of the mission space. Below it, where range requirements are short, a multirotor is cheaper and simpler and pays none of these bills because it never claimed cruise efficiency. Above it, where range requirements are long, a runway-launched fixed-wing aircraft is more efficient and pays none of them because it never claimed vertical capability. The hybrid sits between the two, and the width of that band is set by how much cruise efficiency the architecture had to surrender.

An architecture that surrenders less does not merely perform better inside the band. It widens the band.

4. Proposed configuration

4.1 Overview

The configuration satisfies the zero-bill condition of Section 3.6 directly rather than by compensation. The aircraft stands on its tail. Its entire airframe is a blended-wing body: there is no cylindrical fuselage, and every part of the planform carries payload and produces lift. A single coaxial counter-rotating propeller pair at the nose produces all propulsive thrust, in hover and in cruise alike, without changing its orientation relative to the airframe. Four small coaxial pairs, mounted on rigid frames at the wing tips, produce attitude control and nothing else. The energy system is a series hybrid: fuel drives an

internal-combustion engine, the engine drives a generator, and the generator supplies electric machines at the rotors.

The aircraft has no elevons, no rudder, no tilting mechanism, no retraction mechanism and no dedicated lift system. The only moving aerodynamic device is an on-off strip on the lower surface, described in Section 4.4, which exists solely because roll cannot be produced by propellers alone. Figure 4 gives three orthogonal views of the light reference design and Figure 5 a general view of the same geometry.

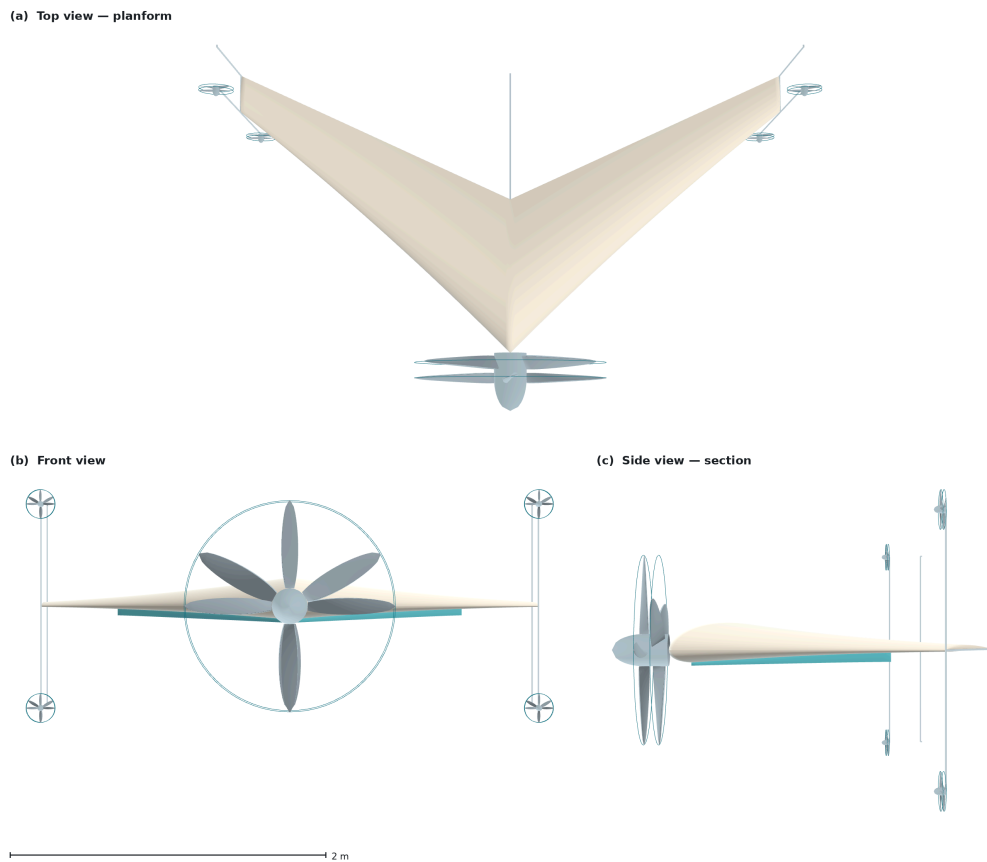


Figure 4. The light reference design, 50 kg, in three orthogonal views: (a) planform, (b) front view, (c) side view showing the aerofoil section and the tip frames. Rendered from the parametric geometry model; the scale bar applies to all three views.

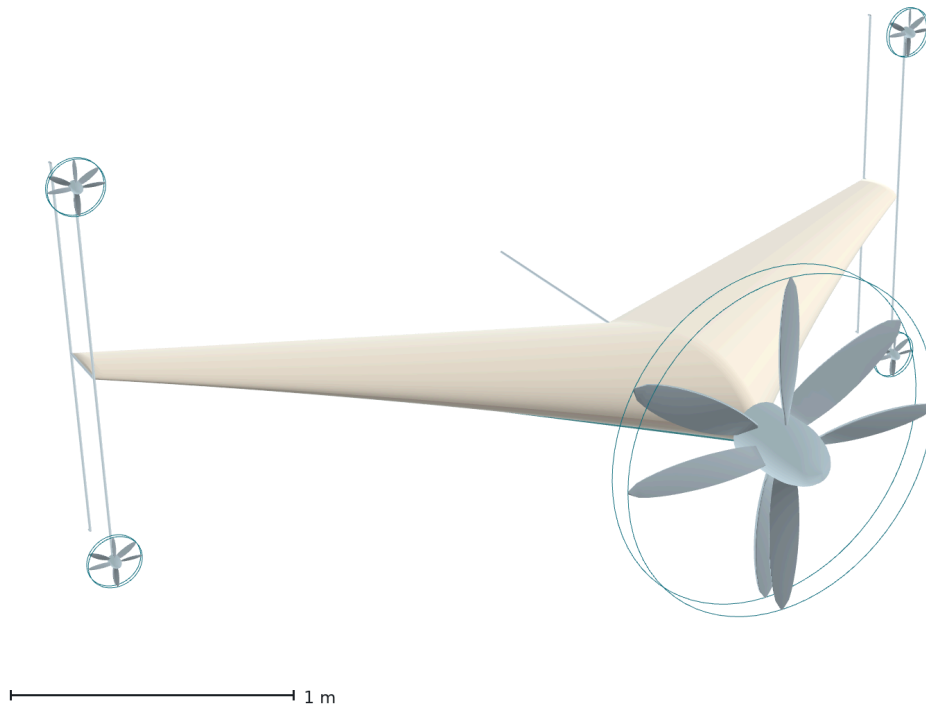


Figure 5. General view of the same geometry. The single coaxial pair at the nose, the four small coaxial pairs at the ends of the tip frames, and the absence of any control surface are all visible.

4.2 Planform

The planform is a blended-wing body whose leading-edge sweep varies continuously along the span while the trailing edge is held at a constant angle. The sweep law is linear in span, running from 45° at the root towards 35° at the span station where leading and trailing edges would converge; the wing is cropped at 67 % of that station, so the sweep actually realised runs from **45° at the root to 38.3° at the tip**. The trailing edge is constant at 25° . Thickness runs from 25 % of chord at the root to 12 % at the tip, and the chord from 0.970 m to 0.236 m — a taper ratio of 0.244. These are one distribution rather than four independent ones, in the manner described in Section 2.3.

The realised sweep variation is therefore modest — under seven degrees — and this should be stated plainly rather than dressed up. The crescent character of the planform comes from the curvature of the leading edge and from the divergence between leading and trailing edge angles, not from a large change in sweep across the span.

These values were chosen, not derived. The 20° – 40° band from which they started was reported favourable in the transonic-transport literature from which the crescent-wing idea comes, and the present aircraft is subsonic, so that band does not transfer on its own authority. The choice is stated here as a design decision and is listed again in Section 8 among the limitations.

Sweep does two jobs in this aircraft, and the second is the reason it is not a free parameter. The first is the conventional one. The second is that the aircraft is tailless: with

no horizontal stabiliser on a boom, the pitching moment must be generated by the distribution of lift along the body itself, and sweep is what places the outboard sections behind the centre of gravity so that they can do it. The sweep angle and the longitudinal stability of the aircraft are therefore the same design variable seen from two directions.

Decreasing sweep outboard also keeps the tip sections from being the first to stall. For a tailless aircraft this matters more than usual, because a tip stall on a swept planform moves the centre of pressure forward and pitches the aircraft further into the stall, and there is no tail with which to argue.

The reference geometry for the light design point is a root chord of 0.970 m, a tip chord of 0.236 m, a span of 3.453 m, a wing area of 1.979 m² and an aspect ratio of 6.03, giving a wing loading of 25.3 kg m⁻² and a stall speed of 20.1 m s⁻¹ against a cruise speed of 30 m s⁻¹. These follow from the sweep and crop laws above rather than being specified independently; Figure 6 gives the distributions.

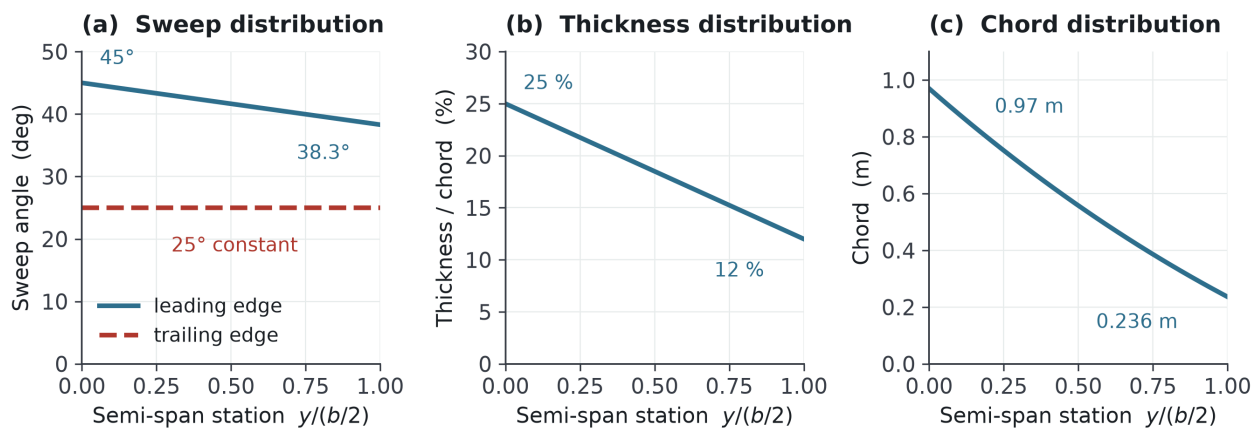


Figure 6. Spanwise distributions of the planform. (a) Leading-edge sweep falls from 45° at the root to 38.3° at the crop station while the trailing edge is held constant at 25°; (b) thickness-to-chord ratio; (c) chord. These are one coupled distribution rather than three independent choices.

4.3 Propulsion

Every propeller on the aircraft is a coaxial counter-rotating pair. There is exactly one reason for this, and it is worth stating narrowly because coaxial rotors are often adopted for other reasons that do not apply here. The reason is reaction torque. A single propeller applies to the airframe a torque equal and opposite to the one it applies to the air. In a tail-sitter that torque acts about the vertical axis in hover and about the roll axis in cruise, and in both cases it must be opposed continuously by something. Opposing it with a control surface costs drag; opposing it with differential thrust costs a control channel. A counter-rotating pair does not produce it.

The pairs are of fixed geometry. The two rotors of a pair may carry different blade twist, but the twist is fixed, and there is no cyclic pitch, no collective pitch and no variable mechanism of any kind. The design point at which torque balance is exact is cruise, not hover; a small residual torque therefore remains in hover, and Section 8 records this.

The counter-rotating arrangement has a second consequence that the transition analysis depends on. Because the two rotors of each pair carry equal and opposite angular momentum, the net angular momentum of the propulsion system is nominally zero. Rotating the airframe through ninety degrees during transition therefore does not precess anything, and no gyroscopic moment appears that the control system would have to cancel. In a tilting architecture this term is present and must be designed for; here it is absent by construction.

Crucially, the pair is never a contra-rotating gearbox. Each rotor is driven by its own electric machine on a common axis. The mechanism that repeatedly defeated the XB-35 — concentric shafts, a splitting gearbox, and the governors that synchronise them — is never built. This is a direct benefit of electric drive and is one of the three things listed in Section 2.5 as available now and not in 1954.

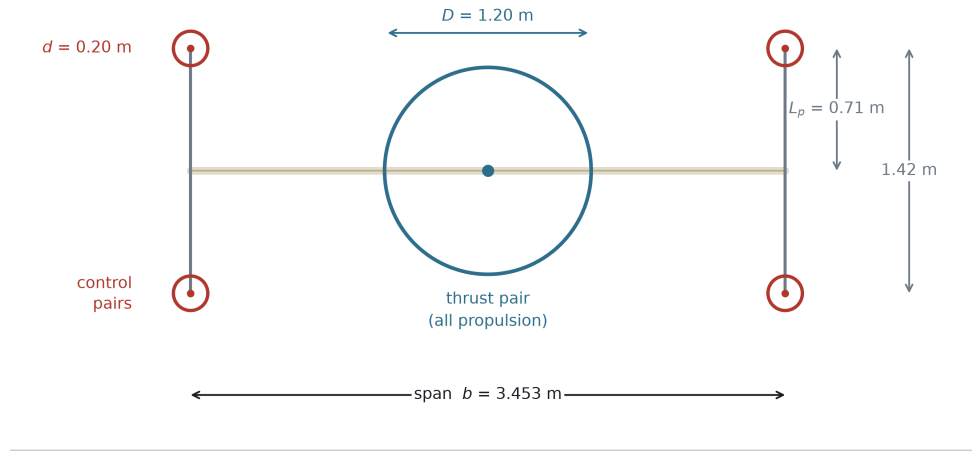
The energy path is a series hybrid: fuel → internal-combustion engine → generator → electric machines. The engine is not mechanically connected to any rotor; it is an energy source. This decoupling is what allows the engine to be sized for cruise rather than for hover. For the light reference design the continuous cruise requirement is 1.9 kW at the engine shaft — 1.7 kW at the electric machines — and the engine is sized at 2.6 kW, while the hover requirement is 10.9 kW at the rotor; the difference is supplied for the duration of the vertical phase by a battery buffer of 1.8 kg, which is 3.6 percent of the maximum take-off mass. Section 5 returns to this, because it is the mechanism by which Bill 3 is not paid.

4.4 Control without control surfaces

The nose pair produces thrust. The four tip pairs produce moments. They are not lift rotors and they are not sized to hover the aircraft; in the vertical phase they carry under fifteen percent of the total power. For the light reference design each tip pair is 0.20 m in diameter and produces 16.2 N during the transition manoeuvre, drawing 335 W, for a total of 1.34 kW across the four.

The tip pairs sit at the ends of rigid frames that extend from each wing tip perpendicular to the planform, above and below, by three hundred percent of the local tip chord — 0.71 m in each direction, giving a vertical separation of 1.42 m between the upper and lower pairs. Figure 7 gives the placement and the resulting moment arms. The frames are long on purpose. The control moment is $M = 2 T L$, so lengthening the arm buys the same moment with less thrust; and because propeller power goes as thrust to the three-halves power, tripling the arm reduces the power required for a given moment to roughly one fifth. The frames are structure that is already needed for another reason, as Section 4.5 explains, so the arm is nearly free.

Propeller placement and moment arms — front view



All thrust vectors are parallel to the body x axis: $\mathbf{F} = (F_x, 0, 0)$

pitch	$M_y = z F_x$	upper vs lower pairs
yaw	$M_z = -y F_x$	left vs right pairs
roll	$M_x = y F_z - z F_y = 0$	identically, at every thrust setting

Figure 7. Propeller placement and moment arms, front view. Every thrust vector is parallel to the body x axis. Pitch and yaw follow directly from differential thrust; the rolling moment is identically zero at every thrust setting, which is the gap the strip of Figure 8 exists to close.

Pitch and yaw follow immediately. Differential thrust between the upper and lower pairs produces a moment about one lateral axis; differential thrust between the left and right pairs produces a moment about the other. In hover these are the two axes the aircraft must control against disturbance; in cruise, with the airframe rotated through ninety degrees, the same four actuators address the same two axes with their roles exchanged. No actuator changes its function, its orientation, or its mounting.

Roll is different, and this is the one place where propellers alone are not sufficient. The result is elementary but decisive. Every propeller on this aircraft has its thrust vector parallel to the body's longitudinal axis, so each thrust is a force $\mathbf{F} = (F_x, 0, 0)$ applied at a station $\mathbf{r} = (x, y, z)$. The moment about the longitudinal axis is

$$M_x = y F_z - z F_y = 0$$

identically, for every propeller, at every thrust setting, regardless of where it is mounted. No arrangement of parallel thrust vectors, and no number of them, can produce a rolling moment. The only roll moment available from the propulsion system is the residual reaction torque, which the counter-rotating arrangement of Section 4.3 has deliberately reduced to nearly zero. The two design decisions oppose one another, and the opposition is real rather than apparent.

This is the same limitation that constrains every tail-sitter, and it is worth being precise about how others resolve it, because the resolution here is not a variation on theirs but a consequence of a different decision made earlier.

Contemporary tail-sitters resolve it in one of two ways. Some place elevons in the propeller wash, which is a control surface by any definition. Others — including a carbon-fibre quadrotor tail-sitter that achieves full attitude control in every flight mode with no control surfaces at all — carry four separate single-rotation propellers and take their rolling moment from the *differential reaction torque* between them. That is the same mechanism a multicopter uses for yaw, appearing as roll once the airframe is rotated into wing-borne flight.

The second solution is unavailable here, and unavailable by construction. The coaxial counter-rotating arrangement of Section 4.3 exists precisely to cancel reaction torque, and it cancels the roll actuator along with it. The two decisions are not merely in tension, as noted above; they are mutually exclusive. A tail-sitter cannot both null its reaction torque and use that torque to roll.

This is the fork at which the present configuration departs from its nearest relatives, and it is why the strip is not an accessory. It is the element that makes the combination possible. The resolution adopted here is different in kind. A single strip on the lower surface, inclined at forty-five degrees, deploys on or off — it is not a proportional control surface. For the light reference design it runs one hundred and twenty percent of the root chord in length, reaching outboard to sixty-seven percent of the semi-span, and stands 2 cm high at its inboard end and 6 cm at its outboard end.

Its authority comes from its length, not its height. The moment scales with the moment arm, whereas the benefit of additional height saturates: at constant length, doubling the height from 5 cm to 10 cm roughly doubles the roll rate, while extending the length from sixty to one hundred and twenty percent of root chord raises it almost fourfold. The estimated roll moment is approximately 46 N·m, giving a roll rate of the order of twenty to twenty-five degrees per second, or 1.2 to 1.5 seconds to a thirty-degree bank.

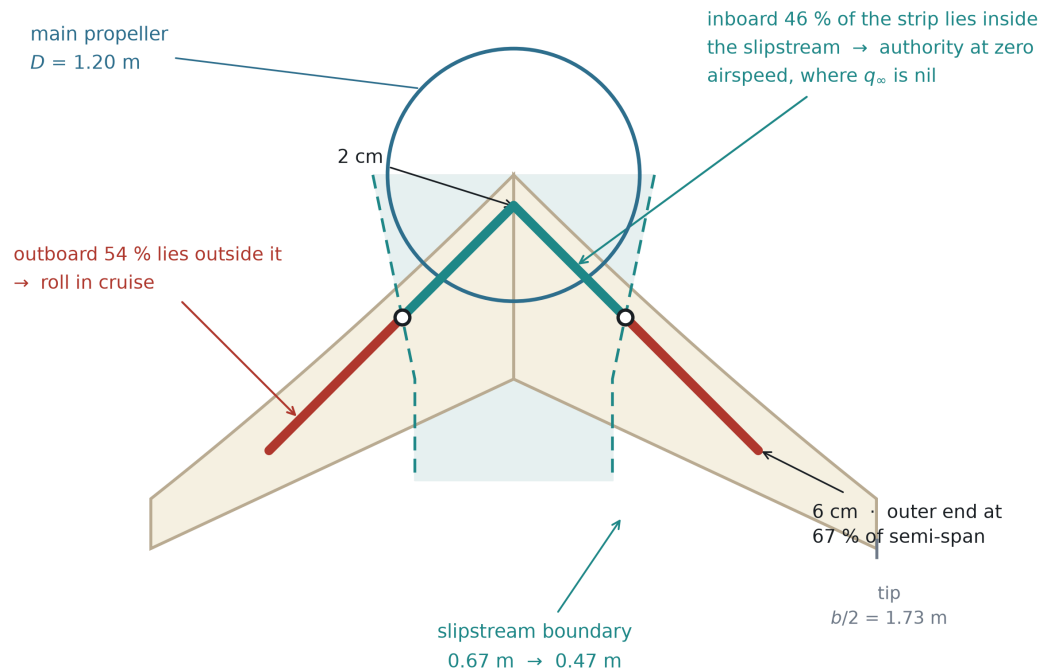
These figures are order-of-magnitude estimates: the damping and control-effectiveness coefficients are taken from the literature rather than computed for this geometry. The length-versus-height conclusion is robust to that choice, because the proportional difference between the two is far larger than the uncertainty in the coefficients.

The strip does one further thing that an ordinary aerodynamic surface cannot. Its inboard portion lies inside the slipstream of the nose propeller, where the dynamic pressure is set by the disc loading rather than by the airspeed:

$$q_{\text{slipstream}} = T / A$$

For the light reference design this is 433 N m⁻² in hover — the dynamic pressure of a 26 m s⁻¹ freestream — available at zero airspeed. The strip therefore produces a usable moment while the aircraft is standing still, which an aerodynamic surface outside the slipstream cannot. Its outboard portion lies beyond the slipstream, where it works against the freestream in cruise. The slipstream covers only twenty-seven to thirty-nine percent of the semi-span, so lengthening the strip to serve cruise does not compromise its hover function; one device serves two regimes. Figure 8 shows the strip against the slipstream boundary: the inboard 46 % of its length lies inside, the outboard 54 % outside.

Roll strip and the main-propeller slipstream — view from below



One device, two regimes. The strip is on the lower surface, inclined at 45° , and deploys on-off. In hover the slipstream supplies the dynamic pressure ($q = T/A = 433 \text{ Pa}$) that the freestream cannot.

Figure 8. The roll strip against the main-propeller slipstream, viewed from below. The inboard 46 % of the strip lies inside the slipstream, where the dynamic pressure is set by disc loading and is therefore available at zero airspeed; the outboard 54 % lies outside it and works against the freestream in cruise.

4.5 Structure and ground contact

The tip frames are not added for the propellers. They are the landing structure.

This inverts a cost into a saving, and the inversion has precedent. Reviewing the tail-sitters of the 1950s, NASA noted that "dispensing with a conventional landing gear improved the empty weight fraction for these VATOL aircraft", while adding that some form of gear was still required on the tail surfaces [2]. The present configuration takes the same benefit and extends it: the structure that meets the ground is also the structure that carries the control propellers and sets their moment arm.

The aircraft rests on five points: the four lower ends of the tip frames and a single keel that runs aft along the centreline from the nose propeller to the trailing edge. Because the aircraft stands on its tail, these five points are what it stands on, and their spread is the stance base that resists tipping in wind. Lengthening the frames to buy the control moment arm of Section 4.4 widens the stance base at the same time. One structure serves three purposes — mounting the control propellers, providing the moment arm, and carrying the landing loads — and is charged to the mass budget once.

The stance base is a design parameter, not a constraint imposed by the configuration. Moving the frame ends further outboard widens it without altering the planform, the propulsion, or the control architecture — and because the same displacement lengthens

the control moment arm of Section 4.4, the two benefits arrive together from one change. The reference geometry given here is one point on that trade; an operator with a stronger ground-wind requirement can take another without redesigning the aircraft.

5. How the architectural tax is avoided

Section 3 identified three bills and argued that they are one quantity paid in three currencies. Section 4 described a configuration built to satisfy the zero-bill condition. This section audits that claim bill by bill, and then states, in the same detail, what the configuration does pay. The second half is not a concession appended for balance. An architecture that claimed to pay nothing would be describing a different aircraft from the one in Section 4.

5.1 Bill 1 — mass: not paid

There is no second propulsion group. The nose pair that lifts the aircraft off the ground is the same pair, in the same orientation, at the same station, that propels it in cruise. Nothing is carried unused.

This is the whole of the argument, and its brevity is the point. The bill was never a consequence of poor design in lift-plus-cruise aircraft; it was a consequence of counting two propulsion systems where the mission needs one. A configuration that counts one does not reduce the bill — it does not generate it.

The tip pairs are a genuine addition and are accounted for in Section 5.4. They are not a second propulsion group: they are sized for moments rather than for weight, and in the vertical phase they draw 1.34 kW against the nose pair's 10.9 kW, which is twelve percent.

5.2 Bill 2 — drag: mostly not paid

In cruise there is no stopped rotor in the airstream, because there is no rotor that stops. The nose pair is the cruise propulsor and runs at its design condition throughout. The quarter of lift-to-drag ratio that Section 3.3 reports as the measured cost of installing hover hardware is not incurred here — not reduced, not mitigated, but absent, because the hardware that causes it does not exist in this configuration. Nor is there a retraction mechanism, so the transfer of Bill 2 into Bill 1 identified in Section 3.3 does not occur either.

It is worth noting what this avoidance also spares. Where lift rotors are retained, keeping their drag small depends on stopping them at a favourable azimuth, which needs an indexing mechanism; where they are stowed, it needs a retraction mechanism. Both are mass, and both are failure modes. A configuration with no rotor to stop needs neither.

The word "mostly" in this heading is deliberate. The configuration does place hardware in the cruise airstream: the four tip frames and the four control propellers they carry. That is a real payment against Bill 2, and quantifying it produced the single most consequential sizing result in this study.

The frames present 2.84 m of exposed length to the flow, oriented perpendicular to it, for the light reference design. At the cruise dynamic pressure of 551 Pa, against a total cruise drag of 38.6 N, their contribution depends almost entirely on their cross-section:

Frame cross-section	C _D	Share of total cruise drag
Circular tube, 20 mm	1.15	93 %
Faired strut, 20 mm	0.15	12 %
Well-faired strut, 20 mm	0.08	6.5 %

Left as circular tubing, the frames alone would produce very nearly as much drag as the entire rest of the aircraft, and the configuration's central claim would collapse. The result is therefore not an observation but a requirement: **the tip frames must be faired**. With a faired section the payment is real and affordable — of the order of twelve percent of cruise drag — and it is reported here as a cost rather than absorbed silently.

One property of this cost is worth noting. The frame frontal area scales with the square of length, and so does the wing area, so under geometric scaling at equal cruise dynamic pressure the fraction is preserved. This bill does not grow with the aircraft.

The drag coefficients used here are representative values for circular and faired sections at the relevant Reynolds number, and the frame cross-section has not yet been selected. The requirement to fair the frames is robust to that choice — the difference between a circular tube and a faired strut is not a matter of coefficient precision — but the twelve-percent figure is an estimate.

5.3 Bill 3 — power system sizing: not paid

The series-hybrid arrangement of Section 4.3 breaks the link that forces the power system to be sized by the hover condition. Because the engine drives a generator rather than a rotor, it supplies average power, not peak power, and the peak is supplied from a buffer.

For the light reference design the numbers are as follows. Cruise draws 1.7 kW at the electric machines, which is 1.9 kW at the engine shaft once the generator and power electronics are accounted for, and the engine is sized at 2.6 kW. Hover requires 10.9 kW at the rotor — 4.2 times the engine's rating. The difference is drawn for the duration of the vertical phase from a 1.8 kg battery, which is 3.6 percent of the maximum take-off mass. The heavy reference design sits on the same line: 39.2 kW electrical in cruise, 54.3 kW engine, 216.2 kW hover, 40 kg of battery at 4.0 percent of MTOW.

An aircraft of this class whose powerplant had to be sized for hover would carry an engine rated above 10.9 kW instead of 2.6 kW. The mass difference is not recovered elsewhere; it is simply not incurred. That the buffer costs under four percent of MTOW at both design points, twenty times apart in mass, is the numerical statement that this avoidance is architectural rather than a fortunate coincidence of one size.

5.4 What is paid

The honest ledger has four entries.

The control propellers. Four pairs, their motors, mounts and wiring exist only to produce moments. In the vertical phase they draw twelve percent of the power the nose pair draws. This is the configuration's substitute for elevons and a rudder, and it is not free — it is merely cheaper than a second lift system, and it does not sit in the cruise airstream in the way a lift rotor does.

The tip frames. As established in Section 5.2, of the order of twelve percent of cruise drag, conditional on being faired. This is the largest single payment the configuration makes, and it is the price of the moment arm, the propeller mounting and the landing structure combined into one member.

The roll strip. The one moving aerodynamic device on the aircraft. Its cost when retracted is a surface discontinuity; when deployed it is a drag device by construction, but it is deployed only while a roll is being commanded.

The transition manoeuvre. The aircraft must rotate through ninety degrees, and the rotation costs time, horizontal displacement and control power. Section 7 treats it in full and shows that the altitude cost, which is the one usually assumed to dominate, can be brought to zero: rotating slowly and entering the rotation while still climbing removes it entirely at both design points. What remains is not free — the manoeuvre occupies seconds during which the aircraft is neither hovering nor cruising — but it is smaller than the literature on tail-sitters would suggest, and it is the one payment on this list that gets *cheaper* the less it is hurried.

Set against the bills of Section 3, the ledger is favourable but not empty. The configuration does not escape physics; it declines a particular trade. What it pays instead is smaller, and — this is the part that matters for scaling — it does not grow faster than the aircraft.

6. Reference designs at two scales

A configuration argument is only as good as its willingness to become a number. This section sizes two aircraft from the arrangement of Section 4 — one at 50 kg and one at 1000 kg, a factor of twenty apart in mass — using the same equations, the same assumptions and the same architecture. The two points are not a light version and a heavy version of different aircraft. They are the same aircraft at two sizes, and the purpose of presenting both is to show that the proportions hold.

Every number below is calculated, not measured. Section 8 says what that means.

6.1 Sizing method

The method is deliberately elementary, and the equations are given so that any result in this section can be checked by hand.

Hover. Thrust equals weight, and induced power follows from momentum theory:

$$v_i = \sqrt{(T / 2\rho A)} \quad P_i = T^{1.5} / \sqrt{(2\rho A)}$$

with the disc loading $DL = T/A$ as the governing parameter. Figure of merit is applied to obtain shaft power.

Cruise. Lift equals weight, which fixes the lift coefficient at the chosen cruise speed; drag then follows from the polar, and the lift-to-drag ratio is the ratio of the two at that point:

$$C_L = W / (q S), \quad q = \frac{1}{2} \rho V^2$$

$$C_D = C_{D0} + C_L^2 / (\pi \cdot AR \cdot e)$$

$$L/D = C_L / C_D$$

This is evaluated **at the cruise condition**, not at the aircraft's best point. The familiar expression $L/D_{\max} = 0.5 \sqrt{(\pi \cdot AR \cdot e / C_{D0})}$ gives the *maximum* lift-to-drag ratio, which occurs at one particular lift coefficient and therefore at one particular speed — 25.3 m s⁻¹ for the light design, only 1.26 times its stall speed. Cruising there would leave too little margin, so both reference designs cruise at 1.49 times stall instead and accept the lift-to-drag ratio that this condition gives. Using the maximum value while specifying a different cruise speed would overstate the range, and an earlier version of this paper did exactly that; the figures below are computed at the cruise point.

Range. The series-hybrid chain is stated explicitly rather than folded into a single efficiency, because the result is sensitive to it and a reader should be able to disagree with any single link:

Link	Value
Internal-combustion engine	0.28
Generator	0.90
Power electronics	0.95
Electric machine	0.92
Propeller	0.80
Overall	0.176

Fuel energy is taken as 12.9 kWh kg⁻¹. The engine figure is the one that matters most: 0.28 is representative of a small four-stroke engine at its best operating point, and it is the reason the range figures below are lower than an optimistic estimate would give.

Control. Tip-pair thrust follows from the required angular acceleration, $M = 2 T L = I \alpha$, with the transition manoeuvre as the sizing case.

Assumptions carried throughout: sea-level density; no compressibility; span efficiency e assumed at 0.85; C_{D0} assumed at 0.0248 for the light design, which is generous for a clean blended-wing body and is intended to absorb the tip-frame contribution of Section 5.2 — that contribution is 0.0043, or seventeen percent of the assumed C_{D0} , so the

assumption is self-consistent rather than optimistic. Both coefficients remain assumptions in what follows. Section 6.6 does not replace them; it bounds them by independent calculation, which is a weaker but more honest claim.

6.2 Light reference design — 50 kg

Quantity	Value
Maximum take-off mass	50 kg
Root chord	0.97 m
Tip chord	0.236 m
Span	3.45 m
Wing area	1.98 m ²
Aspect ratio	6.00
Wing loading	25.3 kg m ⁻²
Main propeller diameter	1.20 m
Disc loading	44.2 kg m ⁻²
Tip propeller diameter	0.20 m
Frame post length	0.71 m each direction
Stall speed	20.1 m s ⁻¹
Cruise speed	30 m s ⁻¹ (108 km h ⁻¹)
Cruise L/D	12.0
Hover power	10.9 kW
Cruise power, electrical	1.7 kW
Engine rating	2.6 kW
Battery buffer	1.8 kg (3.6 % MTOW)
Fuel	8 kg
Endurance	14.8 h
Range	1 598 km
Transition time	2 s

The mass budget behind this is a target and not a finding: 30 % structure, 16 % propulsion chain, 4 % battery, 8 % avionics and control, 16 % fuel, leaving 26 % — 13 kg — for payload. Paper aircraft are habitually lighter than the ones that get built, and that margin has not been paid in this table. Section 8 repeats this warning, because it is the single most likely place for these numbers to be wrong.

6.3 Heavy reference design — 1000 kg

Quantity	Value
Maximum take-off mass	1000 kg
Root chord	3.25 m
Span	11.55 m
Wing area	22.24 m ²
Wing loading	45.0 kg m ⁻²
Main propeller diameter	5.40 m
Disc loading	43.7 kg m ⁻²
Tip propeller diameter	0.67 m
Frame post length	2.38 m each direction
Cruise speed	40 m s ⁻¹ (144 km h ⁻¹)
Cruise L/D	13.6
Hover power	216.2 kW
Cruise power, electrical	39.2 kW
Engine rating	54.3 kW
Battery buffer	40 kg (4.0 % MTOW)
Fuel	160 kg
Endurance	12.6 h
Range	1 814 km
Transition time	4 s

The heavy design has a longer range than the light one despite a shorter endurance. Both effects come from the same source: the larger aircraft cruises faster and, at a higher Reynolds number, achieves a lower zero-lift drag coefficient and therefore a better lift-to-drag ratio. Nothing in the architecture was changed to obtain this.

6.4 Scale behaviour

Five properties of the scaling are worth separating, because three of them are favourable and two are not. Figure 11 shows the two designs at a common scale, and it shows the second of the unfavourable ones directly.

50 kg
span 3.45 m

1000 kg
span 11.55 m · 20x in mass, 3.35x in length

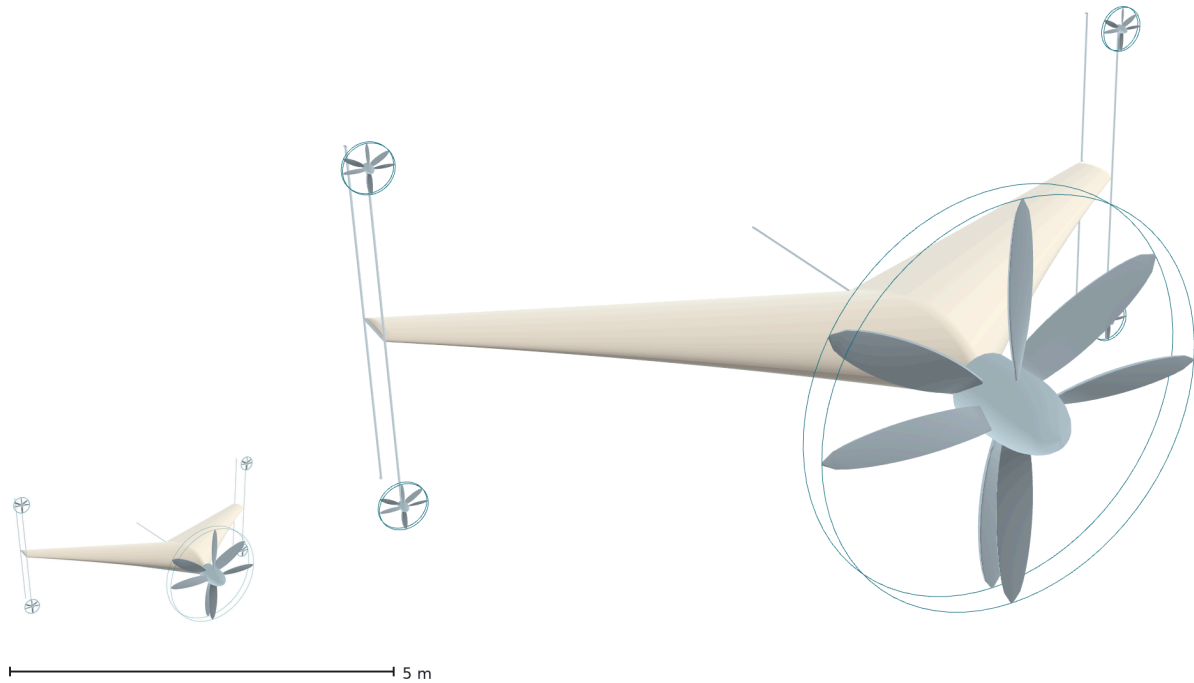


Figure 11. The two reference designs at a common scale: 50 kg and 1000 kg, twenty times apart in mass and 3.35 times in span. Disc loading, energy-buffer mass fraction and tip-frame drag fraction are preserved across that range; transition time is not, and neither is the ratio of propeller diameter to span, which rises from 0.35 to 0.47 because holding the disc loading makes the propeller grow faster than the airframe.

Disc loading is held constant. The two designs sit at 44.2 and 43.7 kg m⁻². This is not a coincidence of sizing but the rule that governs it. Hover power per unit weight is $\sqrt{(DL/2\rho)}$, so holding disc loading constant holds the specific hover power constant, and the aircraft can grow without the hover condition running away. The effect is direct: hover power rises from 10.9 kW to 216.2 kW, a factor of 19.8 against a mass factor of 20. Hover power grows *linearly* with mass rather than as the $L^{3.5}$ of the classical result, and that is the whole benefit of fixing the disc loading.

Constant disc loading means disc area must grow in proportion to weight. For a geometrically similar aircraft, whose mass grows as L^3 , that is a demand for disc area to grow as L^3 rather than as L^2 — which for a fixed number of propellers is impossible. This architecture has two ways out of that, and it uses both. It may add coaxial pairs, and adding a pair is architecturally free because every pair is already torque-balanced on its own; and, as the next paragraph records, it does not hold geometric similarity.

The propeller grows faster than the airframe. This is the second exception, and it is visible in Figure 11 rather than hidden in it. Wing loading is not held constant: it rises from 25.3 to 45.0 kg m⁻², so wing area grows by a factor of 11.2 and span by 3.35 — more than the 2.71 that geometric similarity at constant density would give, but well short of the 4.50 by which the main propeller must grow to hold the disc loading. The ratio of propeller diameter to span therefore rises from 0.35 to 0.47. The heavy design is not the light design photographed from further away; its propeller occupies almost half its span.

Nothing in the argument of this paper fails because of that, since the propeller is the nose of the aircraft rather than an appendage on it, and disc loading — the quantity the hover power depends on — is what is being held. But the claim that the configuration keeps its proportions across the range should be read as applying to the four quantities named here and not to every dimension, and a design much heavier than 1000 kg would reach the point where a single nose pair can no longer hold the disc loading and a second pair must be added.

The buffer fraction is preserved. 3.6 % of MTOW at 50 kg and 4.0 % at 1000 kg. The mechanism by which Bill 3 is avoided therefore does not degrade with size.

The frame drag fraction is preserved. Frontal area and wing area both scale as L^2 , so the twelve percent of Section 5.2 holds at both ends.

Transition time does not scale. This is the exception, and it is stated plainly. The control moment required to rotate the aircraft follows $M = I\alpha$ with $I \propto mL^2$, so the moment needed to turn it in a fixed time grows much faster than the aircraft itself. Scaling the light design's two-second rotation geometrically to 1000 kg would demand 221.5 kW from the tip propellers — 102 % of the hover power, which is to say it is not available at all:

Table 4. Tip-propeller power required to rotate the heavy reference design.

Rotation time	Tip-propeller power, 4 total	Fraction of hover power
2 s	221.5 kW	102 %
3 s	65.6 kW	30 %
4 s	27.7 kW	13 %
5 s	14.2 kW	7 %

The rule is that a larger aircraft turns more slowly. The heavy design rotates in four seconds, at thirteen percent of its hover power.

This constraint is less costly than it first appears, and Section 7 explains why: a slower rotation does not lose more altitude but less, so the scaling penalty on transition time works in the same direction as the scaling penalty on control power rather than against it. The larger aircraft is obliged to turn slowly, and turning slowly is what it should do anyway.

The classical objection to scaling a VTOL aircraft up is that hover power required grows as $L^{3.5}$ while power available grows as L^3 . Fixing the disc loading is what removes that objection on the hover side, as the first item above shows. It does not remove it on the transition side, and Table 4 is where it reappears: the rotation is the one place in this aircraft where the square-cube relation is still paid in full.

6.5 Context

The following aircraft occupy the same mass range. They are listed to locate the reference designs in a real field, not to rank them.

Aircraft	MTOW	Payload	Payload fraction
HAVELSAN BAHA [8]	28 kg	2 kg	7.1 %
Textron Aerosonde Mk 4.7 VTOL [9]	45.4 kg	9.1 kg	20.0 %
Baykar KALKAN [10]	75 kg	~3 kg internal	4.0 %
HAVELSAN BULUT [11]	not published	5 kg	—
Elroy Air Chaparral [12]	865 kg	136 / 227 kg	15.7 / 26.2 %
Sabrewing Rhaegal-A [13]	1400 kg	360–450 kg	25.7–32.1 %
Pipistrel Nuuva V300 [14]	1700 kg	408 kg	24.0 %

All entries are taken from manufacturers' published material. Payload definitions are not consistent between them — some quote internal payload, some external, some both — and empty weights are generally not published. The lightest entry has been confirmed from its manufacturer's data sheet as a vertical take-off aircraft [8].

Three statements can be made about this table and a fourth cannot.

First, the field is real and populated, at both ends of the mass range considered here. Second, payload fraction rises with size across the field, from a few percent at the light end to roughly a quarter at the heavy end, which is the ordinary consequence of fixed costs not scaling down. Third, none of these aircraft connects an internal-combustion engine directly to a lifting rotor; every one of them uses either a generator or separate electric lift, which is independent confirmation that the series arrangement of Section 4.3 is the practical choice at this scale rather than an unusual one.

The fourth statement — that the reference designs outperform these aircraft — is not made, and the numbers in Sections 6.2 and 6.3 should not be read as making it. Those numbers are calculated from a mass budget with an unpaid structural margin; the numbers in this table describe aircraft that exist and fly. Placing a calculation beside a measurement and declaring a winner would be a category error, and the comparison is offered only to show that the reference designs fall inside the field rather than outside it.

A further caution applies to any comparison of endurance or range across this table. Several of these aircraft are fully electric, and for those the endurance figure is set by battery specific energy rather than by configuration. The lightest entry, for instance, is an all-electric fixed-wing VTOL quoting up to two hours of endurance; setting the fuel-burning reference design of Section 6.2 against that number would compare energy sources, not architectures, and would say nothing about the argument of this paper.

What can properly be compared, once these aircraft have been built and flown, is range at similar payload — not payload at similar range. A configuration that carries a comparable load further is making an architectural claim; a configuration that carries a heavier load is making a claim about mass budgeting, which is exactly the part of this study that is least validated.

6.6 Independent checks on the two assumed coefficients

The two coefficients that carry the most weight in Section 6 — span efficiency and zero-lift drag — were assumed rather than derived. They remain assumed. What follows does not replace them with computed values; it asks a narrower question that can be answered honestly: **are the assumed values inside the range that a calculation gives, and on which side?**

Span efficiency. A vortex-lattice solution of the planform of Section 4.2 [15] gives an inviscid span efficiency of 0.99. That is not the same quantity as the 0.85 used here. The vortex-lattice figure counts only the departure of the induced drag from the elliptic ideal; the 0.85 is an Oswald-type efficiency that also carries the viscous drag due to lift, which for a clean wing runs at roughly 85 to 90 percent of the inviscid value. The two are consistent. Reporting the calculation as an improvement on the assumption would be a category error, and it is not claimed.

The same solution gives a lift-curve slope of 3.87 rad^{-1} against the 4.72 rad^{-1} that the transition simulation of Section 7.4 assumes — eighteen percent lower, and in the unfavourable direction. Section 8.6 reports what that does to the transition results.

Zero-lift drag. A strip calculation over the span, taking section drag coefficients at zero lift from a physics-informed aerofoil model [16] and adding the tip frames and the propeller hubs, gives:

Contribution	Light design
Wing and body, clean surface	0.0073
Wing and body, transition tripped near the leading edge	0.0129
Tip frames, faired	0.0043
Tip-propeller hubs	0.0015 – 0.0020
Total	0.0131 – 0.0210

The frame term reproduces the 0.0043 of Section 5.2, which was reached by a different route, and it comes out the same for the heavy design — an independent confirmation of the scale invariance claimed in Section 6.4. The hub term is bounded by hardware rather than guessed: each tip rotor must deliver about 0.83 kgf on a 0.20 m propeller, which places it in the standard 22 mm stator class whose outer cans measure roughly 28 mm.

The assumed 0.0248 lies above the whole of that range. The assumption is therefore conservative in every scenario considered, not merely in the pessimistic one. Because range is linear in lift-to-drag ratio, the reference designs of Sections 6.2 and 6.3 would gain rather than lose if the calculation were adopted — which is the reason it is not adopted. An assumption that is declared and shown to be conservative is a smaller target than a calculation whose weakest link, discussed in Section 8.4, is the treatment of a twenty-five percent thick centre body as a two-dimensional section.

A three-dimensional solution for the centre body. The weakness just named has since been removed. Section 8.12 lists it first among the places these results should be attacked,

and the calculation it asks for has now been carried out: a structured Reynolds-averaged solution over the planform of Section 4.2, at the cruise Reynolds number and at zero lift, resolving the wing and blended body as a three-dimensional surface rather than as stacked sections. It gives a wing-and-body zero-lift drag of

0.01475 with the Spalart-Allmaras closure and 0.01201 - 0.01253 with k- ω SST — a range rather than a single figure, and not for one reason but for two. Nothing in these solutions selects between the two closures, so the eighteen percent between them is carried openly; and the SST value itself is not unique, because two well-converged solutions of the same case from two materially different starting fields settle four percent apart. That second finding is set out below, since it bears on how much weight any of these numbers will hold.

Against the 0.0129 of the tripped strip estimate, the Spalart-Allmaras value is fifteen percent higher and the SST values between two and seven percent lower; the strip method is therefore bracketed rather than simply beaten. Substituting each in turn raises the total to between 0.0203 and 0.0230. **The assumed 0.0248 lies above all of them**, so the conclusion of the previous paragraph survives the more expensive calculation under either closure and under either initialisation; the margin is twenty-two percent on the most optimistic value and eight percent on the conservative one. The assumption is not replaced here either, for the same reason as before. Where a single number is needed downstream, the conservative value is carried.

Neither range is a probabilistic uncertainty band, and neither is reported as one. The remaining terms are measured rather than asserted:

Source of uncertainty	Magnitude
Iterative convergence	0.1 %
Spatial discretisation, three grids of 0.27 - 2.66 M cells	< 0.1 %
Wall resolution, $y^+ 43 \rightarrow 20$	10 %
Wall resolution, $y^+ 20 \rightarrow 1$, Spalart-Allmaras	+ 1.6 %
Wall resolution, $y^+ 20 \rightarrow 1$, k- ω SST	– 6.7 %
Turbulence-model spread, k- ω SST against Spalart-Allmaras, at $y^+ \approx 20$	8 %
Turbulence-model spread, the same pair at $y^+ \approx 1$	18 %
Initialisation spread, k-ω SST at $y^+ \approx 1$, two starting fields	4.3 %

The dominant term is the turbulence model, not the grid — which is worth stating plainly, because grid convergence is the check a reader expects and it turns out to bound the smallest of the terms. The last two rows are labelled *spread* rather than *uncertainty* deliberately: two closures do not sample a distribution, and the interval between them carries no claim that the true value lies inside it. The same caution applies to the last row, which reports two starting fields and not a population of them. Three further results are recorded because they are easy to get wrong in either direction.

The forces converge far more slowly than the residuals: at a velocity residual of 1.6×10^{-5} the computed drag was still fifty-five percent above its converged value, so a solution stopped on residuals alone would have been badly wrong.

The wall-resolution sensitivity does not saturate, and it does not even share a sign between the two models. Resolving the wall raises the Spalart-Allmaras drag by 1.6 percent and lowers the $k-\omega$ SST drag by 6.7 percent, so the spread between the models doubles — from 8 percent at $y^+ \approx 20$ to 18 percent at $y^+ \approx 1$. No single wall-resolution correction is therefore assumed. An earlier version of this section quoted a single value with a five percent band, obtained by averaging one model at $y^+ \approx 1$ against the other at $y^+ \approx 20$; that mixes two wall resolutions and understates the model-form term.

The wall-resolved SST solution is not independent of its starting field. It cannot be started from a uniform field at all: from uniform initial conditions the run develops a localised region of non-physical turbulent kinetic energy near the leading edge at mid-span, which decays over some two thousand iterations without reaching a physical level and then diverges. Two starts that do converge were therefore compared — one warmed from the converged Spalart-Allmaras field, one mapped from the converged SST solution at $y^+ \approx 20$ on the coarser grid. The two initial fields differ by an order of magnitude in peak eddy viscosity and by a factor of six in peak turbulent kinetic energy; the cases are otherwise identical, sharing the same grid file, the same transport and turbulence properties, and the same boundary conditions. Both ran five thousand iterations without bounding, the mapped start reaching the lower velocity residual of the two, 1.0×10^{-7} .

They do not agree. The drag coefficients are 0.01253 and 0.01201, four and three tenths of a percent apart, and the difference lies almost entirely in the pressure component: 0.00386 against 0.00336, thirteen percent, while the viscous components agree to three parts in a thousand. Identical viscous drag with divergent pressure drag places the disagreement in the separated flow rather than in the boundary layer, which is consistent with a steady RANS problem admitting more than one stationary solution — a known possibility in separated flow, but one that removes the uniqueness of the number. A converged steady solution should not depend on its starting point; here it demonstrably does, and the test was run precisely to find out. Both values are therefore reported. Nothing shows that a third starting field would fall inside the interval between them, so 0.01201 – 0.01253 is a measured spread and not a bound.

What this does not settle. The solution is fully turbulent throughout. It therefore speaks to the tripped row of the table above and not to the clean-surface row, and the gap between those two rows — 0.0073 against 0.0120–0.0148 — is now the largest single uncertainty in the zero-lift drag, larger than the spread between the two closures. Closing it requires a transition-sensitive model, which needs a wall-resolved grid and a two-equation formulation at the same time. That combination now converges, and the Langtry-Menter γ - $Re\theta$ model reproduces the ERCOFTAC T3A flat plate to within a few percent in the turbulent region while predicting transition about twenty-five percent early in Reynolds number. But T3A is a bypass case at three percent freestream turbulence, and the present cruise condition is an order of magnitude quieter; attempts to validate the model in that regime were not successful, for reasons traced to leading-edge turbulence

production and freestream decay rather than to the model itself. The clean-surface bound therefore remains where the strip calculation left it, and no transition-model drag figure is claimed.

Internal volume. One closure that the mass budget does not address is whether the payload fits. The body's gross internal volume follows from the thickness distribution: 185 litres for the light design. Taking rather more than half of that as usable and subtracting fuel, buffer, powerplant and avionics leaves about 80 litres for 13 kg of payload, which requires a mean density of only 0.16 kg per litre. The configuration is mass-limited rather than volume-limited, and the choice of mission therefore constrains the structure and the load paths rather than the internal arrangement.

7. Flight profile and transition

The transition between vertical and horizontal flight is the manoeuvre on which tail-sitters have historically been judged, and it is the part of this configuration that most deserves scrutiny. This section describes the flight profile, states the equations that govern the transition, and reports a simulation of it. One result contradicts a widely-assumed relationship and is presented as such.

7.1 The five phases

Stance. The aircraft rests on five points — the four lower ends of the tip frames and the aft end of the centre keel — with its longitudinal axis vertical. No launch equipment is present, and the aircraft is in its own storage attitude.

Vertical take-off. The nose pair spools to a thrust exceeding weight and the aircraft rises vertically. Attitude is held by the four tip pairs. This is the highest-power phase of the flight and the shortest.

Transition. The aircraft rotates from vertical to horizontal while accelerating, until the wing carries the weight. Treated in detail below.

Cruise. The aircraft flies as a tailless blended-wing body. The nose pair is now the cruise propulsor at its design point. The tip pairs provide pitch and yaw, and the lower-surface strip provides roll.

Landing. The reverse of transition, followed by a vertical descent onto the five contact points. Section 7.5 notes what is and is not analysed here.

Figure 9 shows the five phases in sequence. Nothing on the aircraft rotates relative to the aircraft at any point in it.

Flight profile — five phases

Body angle measured from the vertical. The aircraft rotates; nothing on the aircraft rotates relative to it.

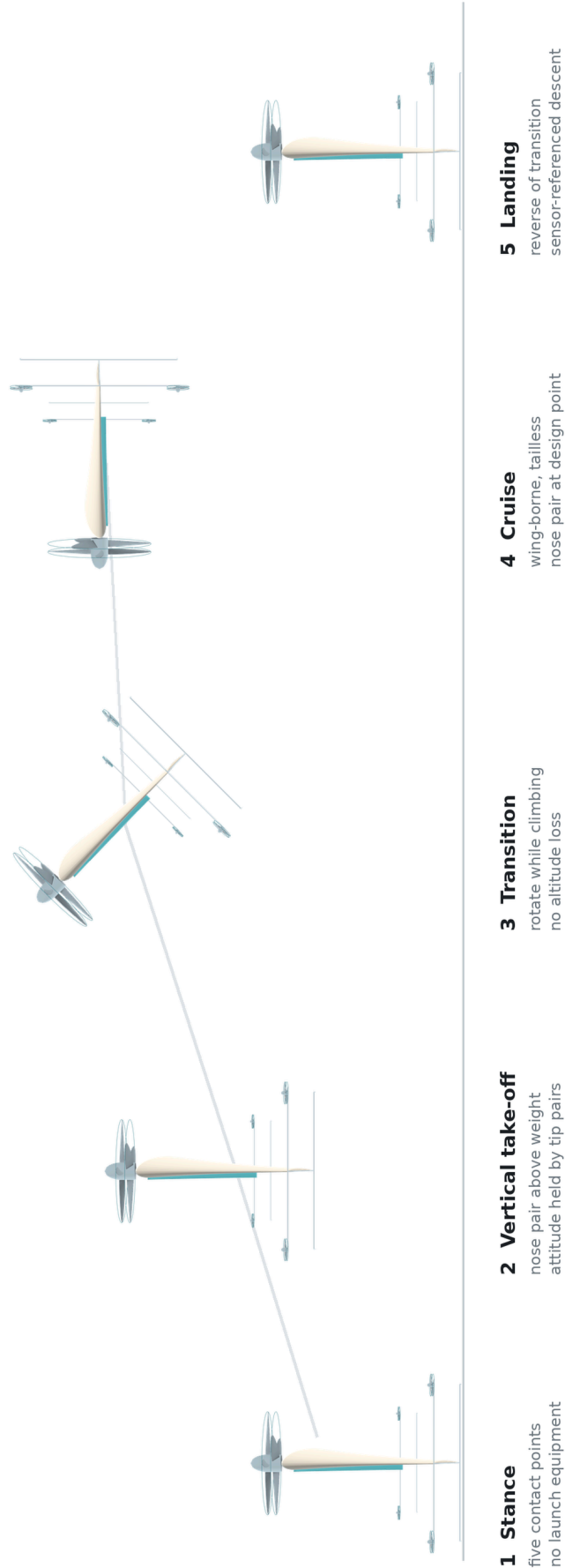


Figure 9. The five phases of the flight profile, drawn from the parametric geometry model. The rotation in phase 3 is treated in Sections 7.3 and 7.4.

7.2 Why the transition begins in the easiest condition

A common objection to tail-sitter transition is that the aircraft must fight the airflow while rotating. For a transition that begins in hover, it does not. At the start of the manoeuvre the airspeed is zero, so the free-stream dynamic pressure

$$q = \frac{1}{2}\rho V^2$$

is zero, and with it every aerodynamic moment that would resist the rotation. The aircraft is not turning against the air; it is turning in still air and then meeting the air as it accelerates.

The consequence is that the difficult part of the transition is not its beginning but its middle, where the airspeed has grown enough for aerodynamic moments to matter but the wing is not yet carrying the weight. The control authority requirement is set there, not at the start.

7.3 The thrust singularity that is never reached

Consider the aircraft at an angle θ from the vertical, and suppose for a moment that it must hold altitude with thrust alone. Vertical equilibrium then requires

$$T \cos \theta = W \quad \rightarrow \quad T = W / \cos \theta$$

which diverges as θ approaches ninety degrees. Read literally, this says a tail-sitter cannot complete a transition, and the expression is sometimes quoted to that effect.

The expression is correct and the conclusion drawn from it is not, because the premise is false. The aircraft does not hold altitude with thrust alone. Vertical support is

$$T \cos \theta + L = W, \quad L = \frac{1}{2}\rho V^2 S C_L$$

and V is not zero during the rotation — it is growing, because the horizontal component $T \sin \theta$ is accelerating the aircraft. The horizontal acceleration, in the same idealisation, is

$$a = g \tan \theta$$

so the very rotation that reduces the vertical component of thrust is what generates the airspeed that replaces it. The singularity is never approached because the wing arrives first.

This is the central mechanism of the manoeuvre, and it also explains the result of the next subsection.

7.4 Transition time: slower is better

The transition was simulated as a two-degree-of-freedom point mass. The body angle θ is driven from zero to ninety degrees over a rotation time t_r ; thrust acts along the body axis, lift perpendicular to the velocity vector and drag opposite to it; the lift curve is linear to stall and a flat-plate relation beyond it; and thrust is reduced to the drag value once cruise speed is reached. Altitude loss is reported as the lowest point of the trajectory relative to the entry altitude.

The result is plotted in Figure 10a for both reference designs and four thrust-to-weight ratios; the tables below give the same values.

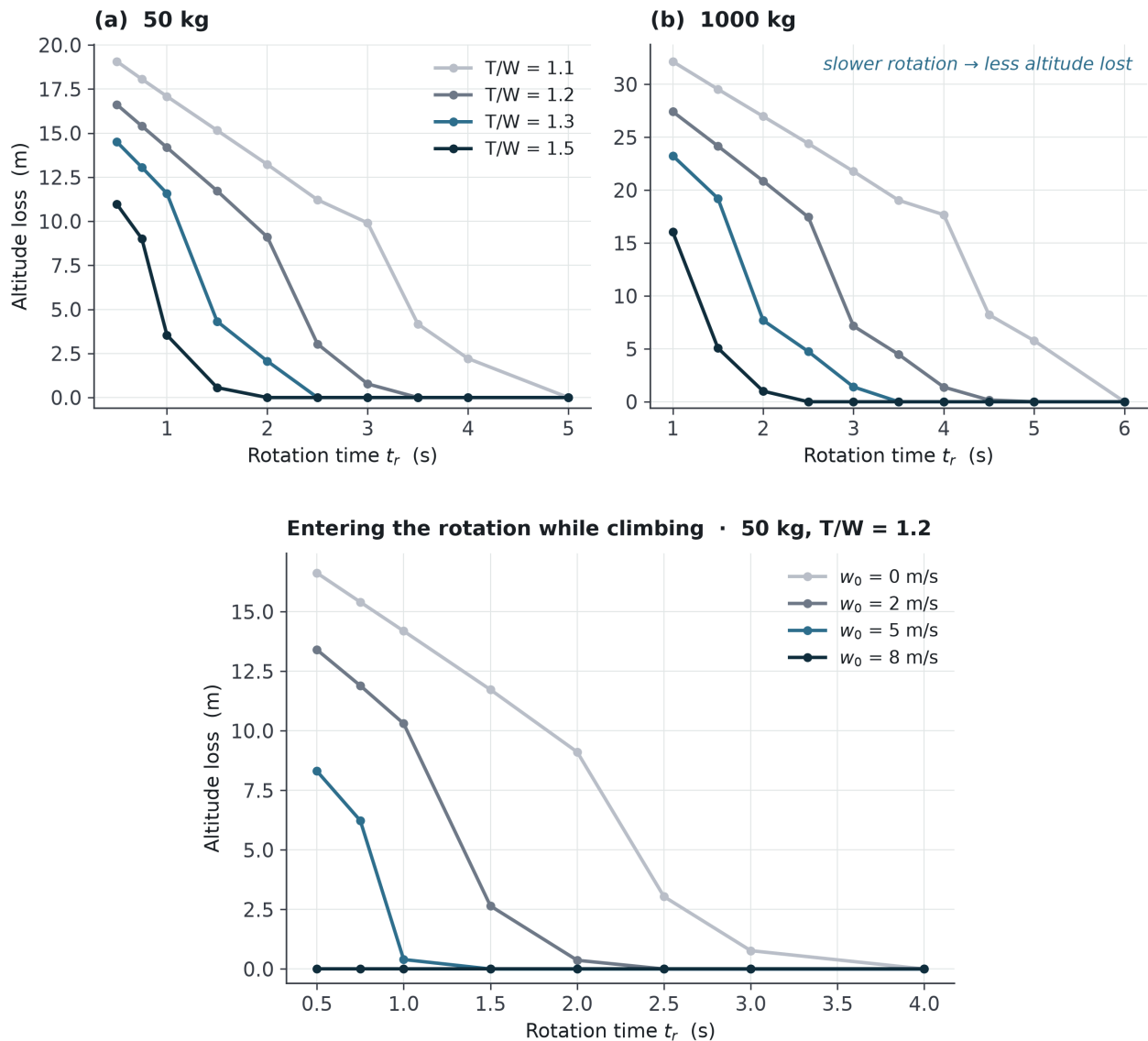


Figure 10. Transition altitude loss. (a) Both reference designs against rotation time at four thrust-to-weight ratios: a slower rotation loses less altitude, not more. (b) The light design at T/W = 1.2 entering the rotation with an upward velocity w_0 ; an entry climb of 5 m s⁻¹ removes the loss at every rotation time considered.

Light reference design, 50 kg. Altitude loss, metres:

t_r	$T/W = 1.1$	$T/W = 1.2$	$T/W = 1.3$	$T/W = 1.5$
0.5 s	-19.1	-16.6	-14.5	-11.0
1 s	-17.1	-14.2	-11.6	-3.5
2 s	-13.2	-9.1	-2.1	0
3 s	-9.9	-0.8	0	0
4 s	-2.2	0	0	0

Heavy reference design, 1000 kg. Altitude loss, metres:

t_r	$T/W = 1.1$	$T/W = 1.2$	$T/W = 1.3$	$T/W = 1.5$
1 s	-32.2	-27.4	-23.3	-16.0
2 s	-27.0	-20.9	-7.7	-1.0
3 s	-21.8	-7.2	-1.4	0
4 s	-17.7	-1.4	0	0
5 s	-5.8	0	0	0

The relationship is monotonic in the direction opposite to the one usually assumed. It is frequently supposed that a tail-sitter should rotate as quickly as possible, on the reasoning that the aircraft is unsupported during the rotation and therefore falls for a time t_r , giving an altitude loss proportional to t_r^2 . **That reasoning is wrong, and the error is in its premise.** The aircraft is not unsupported during the rotation. Vertical support is $T \cos \theta + L$, and a slow rotation keeps $\cos \theta$ large during exactly the interval in which V , and therefore L , is being built. A fast rotation collapses $\cos \theta$ before there is any L to replace it, and the aircraft falls precisely because it hurried.

The practical consequence is a simplification rather than a trade. The control power required to rotate the aircraft in time t_r scales as $1/t_r^2$, so a slow rotation is cheap in control authority; and the altitude loss also falls with t_r . **Both constraints point the same way.** There is no optimum transition time to be found between two competing penalties, because there are not two competing penalties. The transition time is bounded from below, not from above, and its upper bound is set by fuel, horizontal displacement and operational exposure rather than by flight mechanics.

Thrust-to-weight ratio is the dominant parameter. At $T/W = 1.1$ the loss stays in double figures until the rotation is stretched beyond three seconds; at $T/W = 1.3$ the light design reaches zero within three seconds and the heavy design within four. The reference designs assume $T/W = 1.2$, at which the light design completes the manoeuvre in three seconds for a loss under one metre and the heavy design in five seconds for no loss.

Entering the rotation while climbing

The tables above assume the aircraft rotates from a stationary hover. It does not have to, and it should not. The aircraft reaches transition altitude by climbing, which means it

arrives there with an upward velocity that has already been paid for. Stopping to hover before rotating discards that velocity deliberately.

Carrying it into the manoeuvre instead converts it into a reserve. Repeating the simulation with an entry climb rate w_0 :

Light reference design, $T/W = 1.2$. Altitude loss, metres: (plotted in Figure 10b)

t_r	$w_0 = 0$	$w_0 = 2 \text{ m/s}$	$w_0 = 5 \text{ m/s}$	$w_0 = 8 \text{ m/s}$
1 s	-14.2	-10.3	-0.4	0
2 s	-9.1	-0.4	0	0
3 s	-0.8	0	0	0
4 s	0	0	0	0

A five-metre-per-second entry climb removes the altitude loss at every rotation time that is otherwise sensible, and the heavy design behaves the same way. The cost of acquiring that climb rate is negligible: at $T/W = 1.2$ the vertical acceleration is $(T/W - 1)g = 1.96 \text{ m s}^{-2}$, so five metres per second is reached in 2.6 s over 6.4 m of climb, and the kinetic energy involved is 625 J against a fuel energy of 103 kWh.

The reference profile is therefore to enter the rotation at 5 m s^{-1} of climb and rotate over the times given in Section 6 — two seconds for the light design, four for the heavy — for no altitude loss at all. The manoeuvre that the literature treats as the tail-sitter's characteristic hazard becomes, in this configuration, a manoeuvre with no altitude penalty — not because of any device, but because the aircraft is not asked to stop first.

The limits of this simulation should be stated where its results are used. It is a point-mass model in which the body angle is driven kinematically, so it does not represent rotational dynamics and the tip-propeller thrust required to produce the rotation does not follow from it. The aerodynamic model is a linear lift curve to stall with a flat-plate relation beyond it; dynamic stall, separation hysteresis and propeller-wake effects on the wing are absent.

With the transition no longer demanding a rapid rotation, it is no longer the case that sizes the tip propellers. The sizing case becomes disturbance rejection in hover, which an order-of-magnitude estimate places well inside the existing capability: a five-metre-per-second gust normal to the planform produces a moment of the order of 5 N·m, against a capability of $M = 2 T L = 23 \text{ N·m}$ from the tip pairs at the reference geometry — a margin of roughly four. The tip propellers are therefore kept small — 0.20 m — deliberately, so that they contribute as little as possible to cruise drag while retaining margin on the case that actually sizes them.

This is an order-of-magnitude check against a single gust condition. The disturbance spectrum and the closed-loop bandwidth required to reject it have not been analysed.

7.5 Landing

Landing reverses the sequence: the aircraft decelerates, rotates nose-up, and descends vertically onto its five contact points. Two things should be said about it plainly.

The first is that the historical objection to this manoeuvre does not apply. The XFY-1 was cancelled because its pilot had to judge a backwards vertical descent by looking over his shoulder. There is no pilot here, and height above ground is a sensor measurement rather than a human estimate.

The second is that the vertical descent itself has not been analysed in this study. A rotor descending into its own wake can enter the vortex ring state, in which thrust becomes erratic and increasing power makes matters worse. Whether the descent profile of this configuration enters that region, and at what rate of descent, is an open question. It is listed in Section 8 rather than answered here.

8. Limitations

This is a configuration study. It contains no experimental validation of any kind, and the numbers in it are the output of elementary methods applied to a set of assumptions. This section states what those limits are, in enough detail that a reader can judge how much weight each result will bear. Several of the items below were discovered during the study and changed its results; they are recorded here rather than smoothed away.

8.1 No experimental validation

There is no wind-tunnel testing in this work and no flight testing. Nothing in Sections 4 to 7 has been measured against an experiment.

One coefficient has been computed rather than assumed or taken from the literature: the zero-lift drag of the wing and centre body, by a three-dimensional Reynolds-averaged solution reported in Section 6.6. That is a calculation, not a validation — it is subject to the turbulence-model uncertainty quantified there, which is its largest term, and it has not been checked against measurement. Every other aerodynamic coefficient is either taken from the literature or assumed.

8.2 The mass budget is a target, not a finding

The reference designs are sized from an assumed mass breakdown — 30 % structure, 16 % propulsion chain, 4 % battery, 8 % avionics, 16 % fuel, 26 % payload. Paper aircraft are habitually lighter than the aircraft that eventually get built, and no allowance for that margin has been made anywhere in this study. The payload fraction is the number most exposed to it, because payload is the residual: it absorbs the entire error of every other line.

This is the single most likely place for the results of this paper to be wrong, and it is the reason Section 6.5 declines to compare the calculated payload fractions against the

published figures of aircraft that exist.

8.3 Geometry chosen rather than derived

The sweep distribution — 45° at the root falling to 38.3° at the tip on the leading edge, with the trailing edge constant at 25° — was selected from a 20° – 40° band reported favourable in the transonic-transport literature. The present aircraft is subsonic. The band has not been re-derived for this flight regime, and the values are therefore design choices supported by precedent rather than results. The realised sweep variation is under seven degrees, which is smaller than the crescent-wing precedent of Section 2.3 would suggest; the planform inherits the *principle* of a coupled sweep-chord-thickness distribution, not the magnitude of the original.

The taper distribution has likewise not been optimised, and the aerofoil sections are described by thickness, camber and reflex distributions rather than by specific sections.

8.4 Coefficients taken from the literature

Several results depend on coefficients that were not computed for this geometry:

- The **roll authority** of the lower-surface strip — approximately 46 N·m, or twenty to twenty-five degrees per second — uses damping and control-effectiveness coefficients taken from the literature. The order of magnitude is defensible; the value is not. The length-versus-height conclusion is robust to the coefficient choice, because the proportional difference between the two is far larger than the uncertainty.
- The **frame drag** of Section 5.2 uses C_D values representative of circular and faired sections at the relevant Reynolds number. **The frame cross-section has not been selected.** The conclusion that the frames must be faired is robust — the difference between $C_D = 1.15$ and $C_D = 0.15$ is not a matter of coefficient precision — but the twelve-percent figure is an estimate.
- The **zero-lift drag coefficient** of 0.0248 is assumed rather than adopted from a build-up. Section 6.6 reports a component build-up that brackets it at 0.0131 to 0.0210, so the assumed value is conservative. That build-up had a weak link in its largest term — its strip method treated the root section as a two-dimensional aerofoil of twenty-five percent thickness, and the flow over the centre body of a blended-wing body is not two-dimensional. **That link has since been replaced** by a three-dimensional solution, also reported in Section 6.6, which brackets the wing-and-body term between 0.0120 and 0.0148 depending on the turbulence closure and on the starting field, and the total between 0.0203 and 0.0230 — still below the assumed value in every case. What remains uncertain is no longer the dimensionality but, first, the transition state — the solution is fully turbulent, and the clean-surface case is still the strip estimate — and, second, the uniqueness of the solution itself, since the wall-resolved SST case settles four percent apart from two different starting fields. The build-up is reported as a bound on the assumption rather than as a replacement for it.

- **Span efficiency** is assumed at 0.85. A vortex-lattice solution gives an inviscid span efficiency of 0.99 for this planform, which is consistent with the assumed Oswald-type value once the viscous drag due to lift is allowed for, but does not measure the same quantity and is not offered as a correction to it.

8.5 Torque balance holds at one point only

The propeller pairs are of fixed geometry, so exact torque cancellation occurs at one operating condition. That condition was chosen to be cruise, on the grounds that cruise is long and strategic while hover is short and tactical. A small residual torque therefore remains in hover. It is trimmed by the same lower-surface strip that provides roll in cruise, but the trim authority required has not been computed.

8.6 The transition simulation is a point mass

The results of Section 7.4 come from a two-degree-of-freedom point-mass simulation in which the body angle is driven kinematically. It therefore does **not** model rotational dynamics, and the tip-propeller thrust required to produce the rotation does not follow from it. The aerodynamic model is a linear lift curve to stall with a flat-plate relation beyond it; dynamic stall, separation hysteresis and propeller-wake effects on the wing are absent.

The qualitative conclusion — that a slower rotation loses less altitude, and that entering the rotation while climbing removes the loss entirely — depends on the sign of the vertical support term rather than on the details of the aerodynamic model, and is robust. The specific altitude figures are not.

That robustness has since been tested rather than asserted. The simulation takes its lift-curve slope from the thin-aerofoil expression, 4.72 rad^{-1} ; the vortex-lattice solution of Section 6.6 gives 3.87 rad^{-1} for this planform — eighteen percent lower, and in the direction that would make the aircraft fall further. Repeating both tables with the lower value moves no published entry by more than 1.2 m, and the two reference profiles — two seconds for the light design, four for the heavy, entered at 5 m s^{-1} of climb — still lose no altitude at either slope. The conclusion of Section 7.4 survives an eighteen-percent error in the coefficient it rests on.

8.7 The tip-surface benefit is not quantified

The tip frames extend perpendicular to the planform by 0.41 of the semi-span, which is four to eight times the relative height of a conventional winglet. Giving their fairings a lifting section rather than a symmetric one costs no additional part, mass or mechanism, and induced drag is 33.7 % of cruise drag at the reference condition, so the mechanism has something to act on.

No number is claimed for it. The surface is far outside the geometric range for which textbook winglet relations were established, and estimating the benefit properly requires a panel method or CFD. The associated costs — increased root bending moment, and

increased directional stability that the tip propellers must overcome to command yaw — have likewise not been quantified.

8.8 Disturbance rejection in hover

With the transition no longer sizing the tip propellers, hover disturbance rejection becomes the sizing case. Only an order-of-magnitude check has been made against a single gust condition. The disturbance spectrum, the closed-loop bandwidth required, and the actuator response needed to achieve it have not been analysed.

8.9 Ground handling and crosswind

No claim is made that the aircraft resists tipping in arbitrary ground wind, and the historical record gives a specific reason not to make one: the contemporary assessment of the XFY-1 quoted in Section 2.2 records "tip-over tendencies noted when on ground in gusty air" [1]. This is a property of standing an aircraft on its tail and it is inherited here. The stance base is a design parameter that can be widened without altering the configuration, and the reference geometry represents one point on that trade rather than a limit. Operating limits in ground wind are an operational matter, and published data for this class shows that such limits are ordinary rather than exceptional: a fielded fixed-wing VTOL uncrewed aircraft in the same mass range quotes a wind limit of 15 knots for take-off and landing against 25 knots in cruise [8]. A lower ground-wind limit than cruise limit is the normal condition for VTOL aircraft, not a defect peculiar to tail-sitters. The specific limits for this configuration have not been computed here.

8.10 Vertical descent

The vertical descent has not been analysed. A rotor descending into its own wake can enter the vortex ring state, in which thrust becomes erratic and additional power is counterproductive. Whether and at what descent rate this configuration encounters that region is an open question and one of the more important items of future work, because it bears directly on the landing phase.

8.11 Prior art

The novelty claimed in this paper is a combination, and the elements of that combination individually have antecedents. Blended-wing bodies, tail-sitters, coaxial counter-rotating pairs, series-hybrid powertrains and attitude control by differential thrust have all appeared before, separately and in various partial groupings. In particular, full attitude control of a tail-sitter with no control surfaces at all has been demonstrated and published; that aircraft differs from the present configuration in its planform, in distributing thrust across the span, and — decisively — in taking its rolling moment from the differential reaction torque that the coaxial arrangement here removes by design.

The literature underpinning Sections 2 and 3 was read at first hand where the sources could be obtained. Seven were: the two NASA reviews of United States V/STOL

development on which Section 2.2 rests [1,2], the doctoral study from which the drag measurements of Section 3.3 are taken [3], the QuadPlane wind-tunnel characterisation [4], the stationary-lift-propeller drag study [5], the concept-vehicle sizing study quoted in Section 3.2 [6], and the tail-sitter flight-test paper cited in Sections 3.4 and 4.4 [7]. Three further sources were sought and not obtained — the journal version of [3], an earlier conference paper by the authors of [7], and a 2025 forum paper on stopped-rotor drag — and none of them is relied upon for any claim here; where a claim had rested on the last of these, it was removed rather than retained on a summary. No patent claim text was read in the original; the prior-art position stated here is that of an author survey, not of a professional search.

8.12 What would change these conclusions

The results of this paper would be most efficiently attacked in four places, and they are listed so that they can be:

1. **~~A three-dimensional solution for the centre body.~~ Done.** This was the first place to attack, because the strip method of Section 6.6 could not model the flow over a twenty-five percent thick blended centre body. The solution has since been carried out and is reported in Section 6.6: it gives a wing-and-body C_{D0} of 0.01475 with the Spalart-Allmaras closure and 0.01201 – 0.01253 with $k-\omega$ SST — a spread of eighteen percent between the closures that nothing in the solutions resolves — and it leaves the assumed 0.0248 conservative in every case. The dominant term is the turbulence model, and it is larger than first reported because the two models were subsequently paired at the same wall resolution rather than at two different ones. **What it does not settle** is, first, the transition state — the solution is fully turbulent, so the clean-surface figure of 0.0073 is untested and the gap between it and 0.0120–0.0148 is now the largest single uncertainty in the zero-lift drag, larger than the spread between the closures. A transition-sensitive model requires a wall-resolved grid and a two-equation formulation at once; that combination now converges, but the model could not be validated in the low-turbulence regime the cruise condition sits in. **Second, and newly opened, is the uniqueness of the wall-resolved solution itself.** That case cannot be started from a uniform field, and the two starts that do converge — one warmed from the Spalart-Allmaras solution, one mapped from the coarser SST solution — settle 4.3 percent apart, entirely in the pressure component. Steady RANS is admitting more than one stationary solution here. **This item is therefore narrowed rather than closed,** and what replaces it is stated above.
2. **A structural mass estimate** for the airframe and the tip frames, which would test the payload fraction — the weakest number in the study.
3. **A six-degree-of-freedom transition simulation** with rotational dynamics, which would size the tip propellers properly rather than by order of magnitude.
4. **A panel-method analysis of the tip surfaces,** which would either convert Section 8.7 into a quantified benefit or remove it. The vortex-lattice solution of Section 6.6 covers the planform but not the tip surfaces, which remain unquantified.

None of these requires an experiment. The first has been carried out and its result is folded into Section 6.6; the remaining three are within reach of a follow-on study, and the configuration is described in enough detail in Section 4 and Section 6 for another group to attempt any of them independently. The computational setup, the grid-convergence study and the record of what failed along the way are in the repository, so the first item can be re-run and checked rather than taken on trust.

9. Conclusion

Hybrid vertical take-off aircraft pay for their vertical capability, and this paper has argued that the payment is architectural rather than a defect of implementation. It appears in three currencies — the mass of hardware carried but unused, the drag of hardware exposed but inactive, and a power system sized by a condition that holds for about two percent of the flight — and each known architectural move reduces one of them by increasing another. A NASA sizing study of four VTOL architectures reaches the same conclusion from the opposite direction, finding the lift-plus-cruise concept to be the heaviest of those examined while also having the best cruise efficiency, and naming the cause as the empty weight carried for hover.

Stating the tax in that form makes its escape condition explicit: it is charged whenever hover and cruise are served by hardware that is not the same hardware, doing the same job, in the same orientation. The configuration described here satisfies that condition rather than compensating for failing it. The aircraft rotates; nothing on the aircraft rotates relative to it. A single coaxial pair at the nose provides all thrust in both regimes. Four small coaxial pairs at the wing tips provide moments and nothing else, and a strip on the lower surface closes the one gap that propellers cannot — the rolling moment, which parallel thrust vectors cannot produce at any thrust setting or mounting position. There are no elevons, no rudder, no tilting mechanism, no retraction mechanism and no dedicated lift system.

The configuration was sized at 50 kg and at 1000 kg using the same equations and the same architecture, twenty times apart in mass. Three properties hold across that range: disc loading is constant by design, the energy buffer that decouples the engine from the hover peak stays under four percent of take-off mass at both points, and the drag fraction charged to the tip frames is preserved because frontal area and wing area scale together. Holding the disc loading is what keeps hover power growing linearly with mass instead of as the classical $L^{3.5}$. Two quantities do not scale, and both are reported rather than smoothed: the larger aircraft must rotate more slowly, and its propeller grows faster than its span, so the heavy design is not the light design seen from further away.

Two results emerged during the study that changed it. The tip frames, if left as circular tubing, would produce nearly as much drag as the entire rest of the aircraft; fairing them is not an option but a requirement, and it is also what makes their tip surfaces available as lifting surfaces at no additional part or mass. And the transition does not behave as commonly assumed: a slower rotation loses *less* altitude, not more, because the aircraft is supported during the manoeuvre rather than falling through it — so entering the rotation while still climbing, rather than stopping to hover first, removes the altitude penalty entirely.

What this paper offers is a configuration and its numbers, not a validated aircraft. There is no wind-tunnel data here and no flight test; the mass budget is a target that has not paid the margin such budgets usually owe. One of the four analyses that Section 8 lists as tests of these results has been carried out — a three-dimensional solution for the centre body,

which narrowed the zero-lift drag without overturning it — and the other three have not. The claims most exposed are identified in Section 8, and none of the remaining analyses requires an experiment. The configuration is described in enough detail for another group to attempt any of them independently, and that is the outcome this paper is written to invite.

Declarations

Funding. This research received no external funding.

Conflicts of interest. The authors have filed a patent application covering the aircraft configuration described in this paper (Türkpatent application 2026/014570).

Data availability. All data supporting the reported results are contained within the article. The parametric geometry model, the figure-generation scripts and the transition simulation, together with the aerodynamic calculations of Section 6.6 — including the mesh generator, the case setup, the grid-convergence study and the wall-resolution and turbulence-model sensitivity runs behind the computed zero-lift drag — are openly available at <https://github.com/LORDTEK/meryemAircraft>

Acknowledgements. Artificial-intelligence tools were used during the preparation of this work, for literature searching, numerical checking and language editing. All design decisions, engineering judgements and claims presented in this paper are the authors' own, and the authors accept full responsibility for the content.

References

1. Anderson, S. B. *Historical Overview of V/STOL Aircraft Technology*. NASA Technical Memorandum 81280, NASA Ames Research Center, Moffett Field, CA, March 1981.
2. Nelms, W. P.; Anderson, S. B. *V/STOL Concepts in the United States — Past, Present, and Future*. NASA Technical Memorandum 85938 (Report A-9695), NASA Ames Research Center, Moffett Field, CA, April 1984.
3. Bacchini, A. *Electric VTOL Preliminary Design and Wind Tunnel Tests*. Doctoral thesis, Politecnico di Torino, Department of Mechanical and Aerospace Engineering, XXXII cycle, March 2020.
4. Mathur, A.; Atkins, E. *Wind Tunnel Testing and Aerodynamic Characterization of a QuadPlane Uncrewed Aircraft System*. arXiv:2301.12316, 2023.
5. Sahwee, Z.; Mohd Kamal, N. L.; Abdul Hamid, S.; Norhashim, N.; Lotta, N.; Mohd Asri, M. H. Drag Assessment of Vertical Lift Propeller in Forward Flight for Electric Fixed-Wing VTOL Unmanned Aerial Vehicle. *IOP Conference Series: Materials Science and Engineering* **705**, 012007, 2019. doi:10.1088/1757-899X/705/1/012007
6. Silva, C.; Johnson, W.; Antcliff, K. R.; Patterson, M. D. *VTOL Urban Air Mobility Concept Vehicles for Technology Development*. AIAA Paper 2018-3847, 2018 Aviation Technology, Integration, and Operations Conference, Atlanta, GA, June 2018. doi:10.2514/6.2018-3847
7. Xu, W.; Gu, H.; Qing, Y.; Lin, J.; Zhang, F. *Full Attitude Control of an Efficient Quadrotor Tail-sitter VTOL UAV with Flexible Modes*. arXiv:1903.06393, 2019.
8. HAVELSAN. *BAHA — Sub-Cloud Autonomous UAV*. Manufacturer product sheet. Accessed 27 August 2026.

9. Textron Systems. *Aerosonde Mk 4.7 HQ (VTOL)*. Manufacturer product sheet. Accessed 27 August 2026.
10. Baykar. *KALKAN*. Manufacturer product sheet. Accessed 27 August 2026.
11. HAVELSAN. *BULUT*. Manufacturer product sheet. Accessed 27 August 2026.
12. Elroy Air. *Chaparral*. Manufacturer product sheet. Accessed 27 August 2026.
13. Sabrewing Aircraft Company. *Rhaegal-A*. Manufacturer product sheet. Accessed 27 August 2026.
14. Pipistrel. *Nuuva V300*. Manufacturer product sheet. Accessed 27 August 2026.
15. Sharpe, P. D. *AeroSandbox: A Differentiable Framework for Aircraft Design Optimization*. S.M. thesis, Massachusetts Institute of Technology, 2021. Software: <https://github.com/peterdsharpe/AeroSandbox>
16. Sharpe, P. D. *NeuralFoil: An airfoil aerodynamics analysis tool using physics-informed machine learning*. 2023. Software: <https://github.com/peterdsharpe/NeuralFoil>